From Dual-Stream Manifold-Aligned Tokenizer to Latent Sensor-Invariant Consistency Model: Rapid Generative Super-Resolution for Cross-Sensor Remote Sensing Imagery

Yujie Mao, Guojin He, Guizhou Wang, Ranyu Yin, Hongxuan Zhang, and Ziying Chen

***Abstract*—Diffusion models have established state-of-the-art visual realism in remote sensing super-resolution (SR) but suffer from high inference latency. Although accelerated approaches such as latent consistency models (LCMs) enable efficient one-step sampling in a compressed space, their effectiveness in cross-sensor scenarios is hindered by latent manifold misalignment. This limitation arises from the magnified distributional discrepancy within the compressed latent space, which undermines the stability of consistency training and hinders convergence on heterogeneous data. To address this challenge, we propose a unified framework that explicitly decouples cross-sensor SR into sensor-invariant manifold learning and single-step latent residual generation. First, to effectively bridge the domain gap, we introduce the dual-stream Sinkhorn-contrastive masked autoencoder (DSSC-MAE). Functioning as a domain-aligned tokenizer, it employs a novel hierarchical alignment strategy that integrates reconstruction objectives with global Sinkhorn optimal transport (OT) and local patch-wise contrastive constraints. This design aligns heterogeneous distributions within a unified, structure-preserving latent manifold, effectively mitigating sensor-specific discrepancies. Second, building upon this aligned latent space, we propose the latent sensor-invariant residual consistency model (LSIRCM). Leveraging the sensor-invariant latent representation as a structural anchor, LSIRCM reformulates SR as a controllable latent residual generation task. By incorporating cross-attention mechanisms to integrate this aligned context, it utilizes an anchor Diffusion Transformer (AnchorDiT) backbone with a hybrid consistency training strategy to condense the iterative sampling trajectory into a deterministic single-step mapping. Extensive experiments on public and newly constructed datasets demonstrate that our approach achieves a superior efficiency–quality tradeoff. It surpasses the perceptual quality of state-of-the-art diffusion models while maintaining high structural fidelity, matching the inference speed of regression-based networks. Consequently, our framework offers a robust and computationally efficient solution for cross-sensor remote sensing SR tasks. The code will be publicly available at https://github.com/afomyj/LSIRCM**

***Index Terms*—Cross-sensor, deep learning, latent consistency model (LCM), masked autoencoder (MAE), remote sensing, super-resolution (SR).**

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I. Introduction

High-resolution (HR) remote sensing imagery plays a fundamental role in Earth observation and geospatial analysis, enabling a wide range of applications such as fine-grained land cover mapping [1], precision agriculture [2], photovoltaic panel extraction [3], and environmental assessment [4]. However, operational acquisition is constrained by the inherent tradeoff between spatial resolution and spatiotemporal coverage, often limiting large-scale monitoring to coarser resolutions. Given these limitations, computational enhancement of existing observations becomes essential to unlock their full spatial potential. In this context, single-image super-resolution (SR) offers a powerful approach to synthesize high-fidelity details from inherently limited low-resolution (LR) acquisitions [5], [6], [7].

In the field of remote sensing image SR, regression-based frameworks employing convolutional neural networks (CNNs) or Transformers have long been dominant. While effective at minimizing pixel-wise errors, they are inherently limited by the perception–distortion tradeoff [8], often yielding over-smoothed results that lack high-frequency details [9], [10], [11], [12]. To enhance visual realism, generative adversarial networks (GANs) were introduced. However, their practical application is limited by inherent training instability and the tendency to generate hallucinatory artifacts that violate geometric fidelity [13], [14], [15].

Recently, denoising diffusion probabilistic models (DDPMs) have emerged as a promising alternative. By avoiding the mode

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collapse inherent in GANs, they achieve state-of-the-art perceptual quality in reconstructing fine-scale geospatial textures [16], [17]. Consequently, diffusion-based models have demonstrated exceptional potential in recovering intricate spatial details often smoothed out by regression baselines [18], [19].

Nevertheless, the deployment of diffusion models in large-scale Earth observation is hindered by two fundamental bottlenecks. First, the iterative nature of the reverse diffusion process imposes substantial computational costs and latency [20]. Second, a substantial disparity exists between idealized degradation models and real-world sensor heterogeneity. Unlike natural photography, remote sensing acquisition is governed by complex physical mechanisms, including distinct spectral response functions (SRFs) and bidirectional reflectance distribution function (BRDF) effects. These factors introduce nonlinear domain shifts that invalidate synthetic degradation assumptions [21], [22]. This issue is particularly acute for latent-based generative models. Due to high compression ratios, the latent space becomes highly sensitive to such domain shifts, resulting in latent manifold misalignment [23], which fundamentally impedes generative training.

To address the efficiency issue, latent consistency models (LCMs) [24], [25] have been proposed to enable single-step inference. However, applying these models to cross-sensor remote sensing tasks remains challenging. Consistency training aims to enforce self-consistency along the probability flow ordinary differential equation (ODE) trajectory. In cross-sensor scenarios, the significant spectral and spatial discrepancies between sensors create a large distributional gap between the conditional input and the target data. Consequently, when standard tokenizers compress these unaligned distributions without explicit geometric constraints, they tend to project heterogeneous data into poorly aligned latent submanifolds. From the perspective of probability flow ODE dynamics [26], transport between such topologically misaligned distributions often induces highly complex and nonlinear vector fields. This inherent manifold misalignment increases the curvature of the generated trajectories, potentially amplifying local truncation errors significantly during consistency learning. Since learning a consistency mapping for such high-curvature trajectories is inherently difficult in the compressed latent space, the model struggles to converge, often leading to training instability and suboptimal generation results.

Therefore, constructing a sensor-invariant latent space is critical to reduce this trajectory curvature and facilitate training. However, conventional encoders face limitations in this context. Although convolutional variational autoencoders (VAEs) have been widely adopted in latent diffusion frameworks [27], their reliance on convolutional operations inherently prioritizes local feature extraction. Consequently, they often struggle to capture the global semantic dependencies essential for robust cross-sensor alignment. Similarly, masked autoencoders (MAEs) have emerged as effective tokenizers for diffusion models [28], [29], demonstrating strong semantic extraction capabilities through masked modeling. However, standard implementations typically rely on fixed spatial patch correspondence, lacking explicit geometric alignment constraints required to bridge misaligned cross-sensor

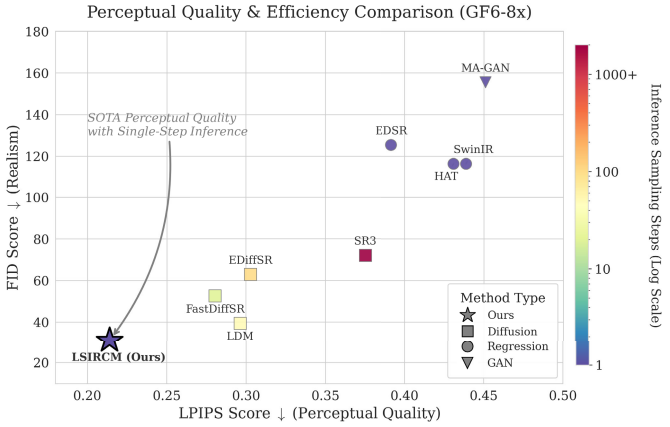


Fig. 1. Perception–efficiency tradeoff on the GF6-8x dataset. LPIPS [39] and FID [15] represent perceptual quality and realism, respectively (lower is better). Marker color denotes sampling steps (log scale). Our LSIRCM (star) achieves an optimal balance, delivering diffusion-level quality with single-step efficiency.

manifolds. To enhance compression efficiency and robustly extract semantic structures from sensor-specific variations, query-driven compression [30], [31], [32] serves as a compelling enhancement to MAE-based tokenizers. Functioning as an active information bottleneck, this paradigm dissolves the strict binding between feature representations and spatial coordinates, effectively facilitating the efficient extraction of consistent high-level semantics across sensors. Furthermore, the adoption of such compressed, sequence-based representations naturally necessitates a shift from convolutional U-Nets to Diffusion Transformers (DiTs) [33], which provide a scalable architecture natively suited for modeling complex distributions within query-compressed latent spaces.

Nevertheless, architectural advancements alone may face challenges in completely resolving complex cross-sensor discrepancies [34]. The substantial cross-sensor heterogeneity highlights the importance of explicitly introducing alignment constraints to rectify distributional mismatches within the latent space.

Driven by these insights, we propose a unified framework that explicitly decouples sensor-invariant manifold learning from controllable residual generation [35], aiming to regularize latent misalignment and straighten the generative trajectory.

Specifically, we introduce the dual-stream Sinkhorn-contrastive MAE (DSSC-MAE). Adopting an asymmetric teacher–student architecture with query-driven compression, it utilizes a masked teacher trained on HR data as a manifold anchor. To align the student's representation, we devise a hierarchical alignment strategy. By integrating reconstruction objectives with local patch-wise contrastive learning [36] and global Sinkhorn optimal transport (OT) [37], [38], this strategy effectively projects heterogeneous distributions into a unified latent manifold. Subsequently, building on this aligned prior, we propose the latent sensor-invariant residual consistency model (LSIRCM). Leveraging an anchor DiT (AnchorDiT) backbone, it reformulates the SR task as predicting fine-grained residuals relative to the prealigned latent anchor. By aligning the latent structures prior to generation, this approach smooths the ODE trajectory, thereby reducing truncation errors

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and promoting stable optimization. To enable high-fidelity single-step inference, we employ a hybrid consistency training strategy augmented by noise-aware weighting, which dynamically balances structural refinement and generative consistency along the straightened probability flow trajectory. As illustrated in Fig. 1, our method exceeds the perceptual realism of state-of-the-art diffusion models while matching the inference speed of deterministic regression-based networks.

In summary, the main contributions of this article are threefold.

1) We establish a unified two-stage framework that explicitly decouples sensor-invariant manifold learning from generative reconstruction, thereby isolating domain heterogeneity to provide a robust solution for cross-sensor remote sensing SR.
2) We propose the DSSC-MAE, which constructs a unified, sensor-invariant latent space to rectify manifold misalignment by implementing a hierarchical alignment strategy that integrates global Sinkhorn OT with local patch-wise contrastive constraints.
3) We introduce the LSIRCM, which leverages an AnchorDiT backbone and hybrid consistency training within the aligned latent space to enable high-fidelity single-step inference, optimizing the tradeoff between generative realism and efficiency.

The remainder of this article is organized as follows. Section II reviews the related work. Section III details the proposed framework. Section IV reports experimental results and ablation studies. Finally, Section V concludes this article.

II. Related Work

A. Deep Learning for Image SR

Single-image SR aims to recover HR details from LR observations. Since the introduction of deep learning, regression-based frameworks have dominated this field. Early CNNs, such as SRCNN [9], VDSR [10], and EDSR [40], focused on learning direct LR-to-HR mappings by minimizing pixel-wise reconstruction errors. These methods established strong benchmarks for distortion-oriented metrics like peak signal-to-noise ratio (PSNR). Recently, Vision Transformers (ViTs) have further improved performance by modeling long-range dependencies. Architectures, including SwinIR [11] and HAT [41], leverage self-attention mechanisms to utilize broader spatial contexts, achieving impressive signal fidelity.

However, methods optimizing pixel-wise objective functions inherently suffer from the regression-to-the-mean phenomenon. By predicting the average of all plausible solutions, they tend to produce oversmoothed results lacking high-frequency textural details, a limitation governed by the perception–distortion tradeoff [8].

To improve perceptual quality, GANs were introduced. Frameworks like SRGAN [42] and ESRGAN [13] employ adversarial loss functions to synthesize photorealistic textures. Nevertheless, GANs face challenges regarding training instability and mode collapse, occasionally producing hallucinatory artifacts [15]. Recently, DDPMs [16], [17] have emerged as a robust alternative. By modeling the data distribution through an iterative denoising process, diffusion models achieve superior perceptual quality. To enhance scalability, recent research has shifted toward latent diffusion models (LDMs) [27] and DiTs [33], which conduct generation within compressed latent spaces to reduce computational complexity and enhance the denoising capability of the latent model.

B. Remote Sensing Image SR

Adapting SR models to remote sensing imagery requires addressing domain-specific challenges, including vast scale variations and complex spatial heterogeneity. Early research focused on adapting CNN architectures to the geospatial domain. For instance, LGCNet [43] introduced local–global mechanisms to learn multiscale representations. To enhance texture realism, GAN-based approaches like MA-GAN [44] utilized multiattention mechanisms to generate visually plausible details. Subsequently, Transformer-based methods such as TTST [45] were employed to capture long-range spatial dependencies effective for large-scale scenes. More recently, diffusion models have demonstrated significant potential in prioritizing perceptual quality. Methods such as EDiffSR [19] and FastDiffSR [46] leverage diffusion priors to reconstruct fine-grained geospatial details often lost in regression-based baselines. Building on this, advanced strategies have emerged, including controllable reference-guided generative priors [47] and residual diffusion models like TTRD3 [48] that directly model high-frequency details. While these residual strategies are highly efficient, their direct application in cross-sensor scenarios is often hampered by domain shifts if the latent space remains misaligned. Our framework explicitly addresses this challenge by aligning the latent representations prior to residual generation, thereby ensuring generative stability under cross-sensor heterogeneity.

A significant limitation across most existing approaches, including these advanced diffusion strategies, is their reliance on synthetic datasets simulated via idealized degradation or single-domain assumptions. Lacking mechanisms to accommodate the complex spectral and spatial variations of real-world sensors, these models often exhibit performance degradation and chaotic residual feature spaces in cross-sensor scenarios [12].

To facilitate research in this domain, the community has introduced large-scale cross-sensor datasets such as SEN2NAIP [22] and OLI2MSI [21], providing paired samples for supervised learning. Building on these resources, recent studies have proposed specialized architectures to handle sensor heterogeneity. DESAT [12] employs strip attention mechanisms to preserve structural integrity, while Miao et al. [49] explore diffusion-based reconstruction for cross-sensor imagery. Furthermore, frameworks focusing on radiometric consistency have been developed to improve physical reliability [34], [50].

C. Generative Acceleration and Adaptation in Remote Sensing

The deployment of generative models in remote sensing is constrained by computational inefficiency. Standard diffusion

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models rely on iterative sampling chains that impose high computational burdens. To mitigate this, research has followed two primary avenues: sampling acceleration and spatial compression. Regarding acceleration, denoising diffusion implicit models (DDIMs) [20] first introduced non-Markovian sampling, treating the reverse process as a deterministic ODE to enable step skipping for faster inference. Pushing this paradigm further, consistency models (CMs) [51], [52] and LCMs [24] distill this probability flow ODE to achieve high-fidelity inference in a single or few steps. Notably, advanced tangent alignment strategies [53] have been shown to further reduce truncation errors and stabilize the sampling trajectory. Concurrently, LDMs [27] compress HR images into a low-dimensional latent space for efficient training. Recent research has begun to explore the application of these combined strategies to remote sensing SR [25], primarily within synthetic scenarios governed by idealized degradation.

However, applying this accelerated paradigm to cross-sensor remote sensing presents a critical challenge. Compressing heterogeneous sensor data into a compact latent space tends to amplify inherent domain discrepancies. This leads to severe manifold misalignment, where uncorrected domain shifts cause the geometric structures of the encoded LR and HR distributions to become disjoint, thereby rendering standard latent training unstable and leading to severe structural distortion during the generative process.

To address this misalignment, establishing a sensor-invariant latent manifold is essential. In the context of representation learning, MAEs have demonstrated to be effective visual tokenizers superior to standard VAEs, extracting semantically rich features that are naturally robust to noise [28]. This semantic emphasis aligns well with recent advances in foundation models, which leverage pretrained knowledge for robust domain adaptation [54] and staged fine-tuning to mitigate data shifts [55]. While recent works [29] have successfully integrated MAEs into latent generative models, these applications are typically designed only for intradomain spatial compression. In complex cross-sensor tasks, relying solely on implicit semantic extraction is often insufficient to bridge misaligned sensor manifolds. From the perspective of probability flow ODE dynamics [26], uncorrected manifold misalignments can cause transport trajectories to intersect, increasing the curvature of generated paths and complicating consistency training.

To further enforce distribution consistency, advanced domain adaptation techniques provide robust mechanisms. Contrastive learning frameworks, such as patch-wise noise contrastive estimation (PatchNCE) [36], which has proven highly effective in unpaired image-to-image translation [56], maximize mutual information between corresponding patches to ensure local alignment. Simultaneously, OT theory, specifically the Sinkhorn algorithm [37], offers a geometric framework to minimize global distributional distances [38]. Consequently, for cross-sensor generative modeling, explicitly aligning the latent representations is a critical prerequisite. By integrating global constraints like Sinkhorn OT with local regularizers like PatchNCE, it is possible to structurally align the heterogeneous inputs prior to generative training. Establishing such a well-aligned, sensor-invariant latent space effectively smooths the ODE trajectory, thereby facilitating stable and efficient single-step inference under cross-sensor heterogeneity.

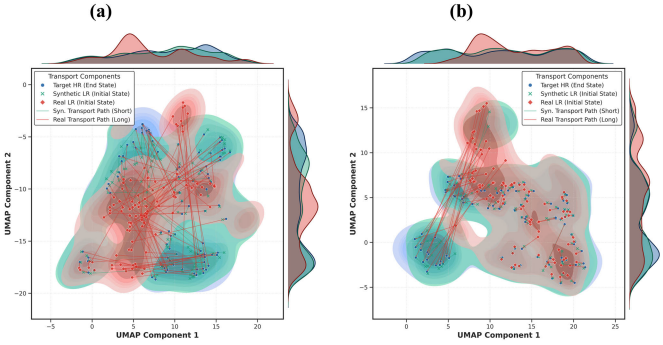


Fig. 2. UMAP visualization of latent manifold misalignments on (a) SEN2NAIP and (b) GF6-8x datasets. In stark contrast to the synthetic baseline (green), real cross-sensor LR states (red) exhibit pronounced spatial displacement from the target HR manifold (blue), suggesting the formation of elongated transport paths across misaligned manifold regions.

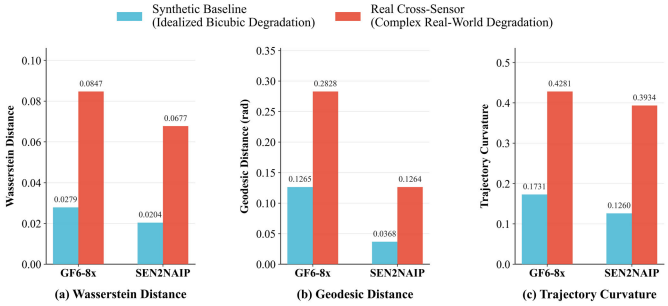


Fig. 3. Quantitative analysis of latent transport dynamics. Real cross-sensor discrepancies significantly elevate (a) manifold transport cost, (b) latent manifold discrepancy, and (c) trajectory curvature. This aligns with our theoretical analysis, demonstrating how uncorrected manifold shifts tend to complicate the generative ODE flow compared to idealized synthetic conditions.

III. Proposed Method

A. Motivation

While LCMs [24] enable efficient single-step inference, their deployment in cross-sensor remote sensing often faces challenges due to severe domain shifts. Unlike idealized synthetic degradations (e.g., bicubic downsampling), real-world cross-sensor images exhibit complex, nonlinear radiometric and geometric variations [57], [58]. When these heterogeneous signals are encoded via standard tokenizers such as KL-regularized variational autoencoders (KL-VAEs), the compressed representations are projected into poorly aligned latent submanifolds, potentially amplifying initial physical discrepancies. From the perspective of OT [37], [38] and probability flow ODE dynamics [26], transport between such misaligned manifolds can induce highly complex, intersecting vector fields. This manifold distortion tends to increase the curvature of the generative trajectory, which may significantly amplify local truncation errors during consistent learning, potentially undermining training stability and inference fidelity.

To empirically explore this bottleneck, we compared the latent manifolds yielded by a standard KL-VAE across synthetic degradation baselines and real cross-sensor scenarios,

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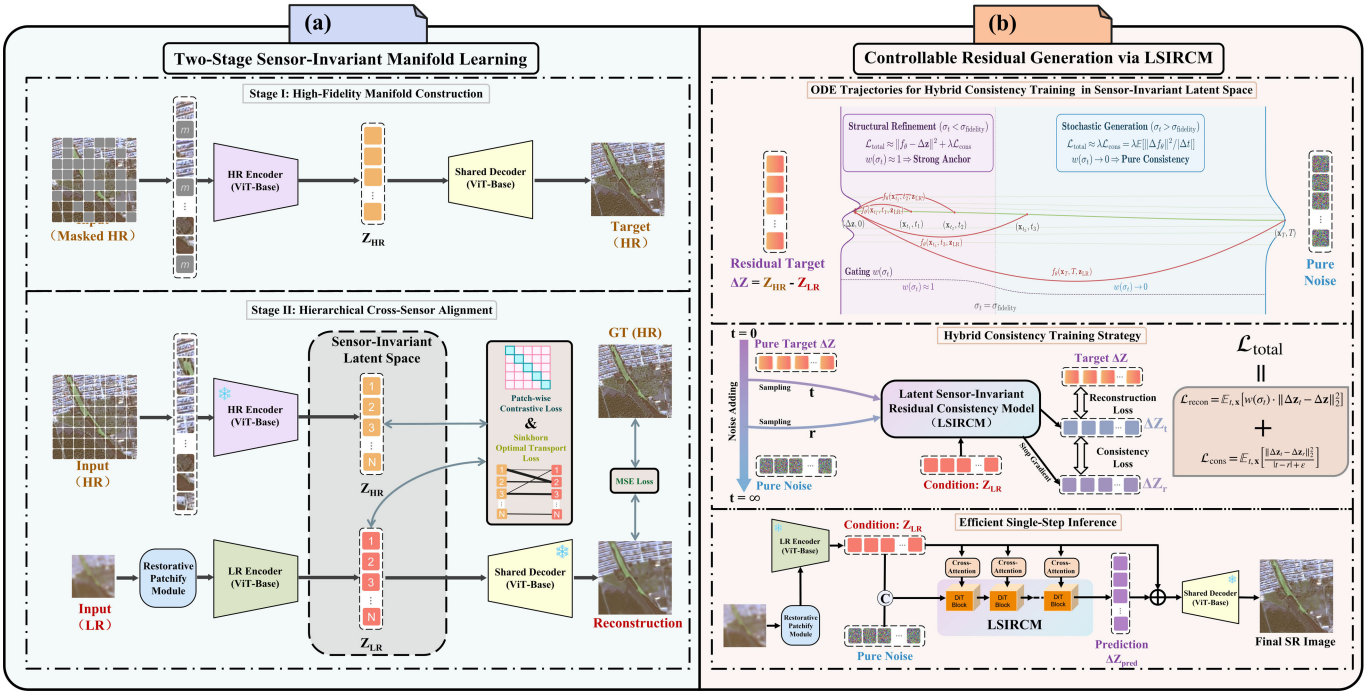


Fig. 4. Overview of the proposed unified framework. (a) Phase 1: two-stage sensor-invariant manifold learning. Stage I constructs a robust target manifold via a masked teacher. Stage II utilizes the RPM and hierarchical constraints (sinkhorn and contrastive) to align the student encoder to this prior, establishing a unified sensor-invariant latent space. (b) Phase 2: controllable residual generation via LSIRCM. Using the aligned latent $\mathbf{Z}_{\mathrm{LR}}$ as a structural anchor, AnchorDiT predicts the residual $\Delta\mathbf{Z}_{\mathrm{pred}}$ via cross-attention. The optimization is modulated by a noise-aware weighting strategy $w(\sigma_t)$ under hybrid consistency training, balancing structural refinement with stochastic generation for deterministic single-step reconstruction.

analyzing how the resulting manifold shifts affect the curvature of the generative ODE trajectories. To ensure a fair comparison of these intrinsic geometric properties, these trajectories were empirically extracted from residual CMs separately optimized for each scenario. As shown in the visualization of UMAP and edge density estimation in Fig. 2, the synthetic LR state (green) is largely aligned with the target HR manifold (blue). In contrast, real cross-sensor LR states (red) exhibit pronounced spatial displacement, often requiring elongated transport paths across misaligned manifold regions. The quantitative analysis in Fig. 3 is consistent with this structural shift: real sensor heterogeneity significantly increases the OT cost (Wasserstein distance) and the domain manifold discrepancy (Gram matrix geodesic distance) compared to the synthetic baseline. Crucially, the empirical trajectory curvature [26] undergoes a substantial increase under real conditions. This evidence suggests that uncorrected manifold shifts tend to complicate the generative ODE flow. Consequently, to facilitate stable single-step generation, it becomes highly beneficial to structurally align these heterogeneous manifolds prior to consistency training. This insight directly motivates our proposed framework: by explicitly decoupling sensor-invariant manifold learning from residual generation, we aim to establish a well-aligned latent manifold that helps smooth the underlying ODE trajectory.

B. Framework Overview

Motivated by the aforementioned geometric insights, we formulate the cross-sensor SR task as a two-stage optimization problem that explicitly decouples sensor-invariant representation learning from generative reconstruction. As illustrated in Fig. 4, the proposed framework addresses the dual challenges of cross-sensor discrepancies and inference latency through two coordinated phases, thereby significantly enhancing the practical feasibility of remote sensing SR.

The first phase, two-stage sensor-invariant manifold learning, bridges the domain gap via a sequential optimization protocol shown in Fig. 4(a). In Stage I, we perform high-fidelity manifold construction to establish a compact and robust target latent space $\mathbf{Z}_{\mathrm{HR}}$ using a masked teacher encoder trained on HR data. Subsequently, Stage II executes hierarchical cross-sensor alignment using the DSSC-MAE. In this stage, a learnable student encoder, equipped with a specialized restorative patchify module (RPM) for high-frequency detail recovery, is fine-tuned to distill knowledge from the frozen teacher. This is achieved through a multidimensional hierarchical alignment strategy that integrates pixel-level fidelity and token-level matching for traditional spatial reconstruction, while synergizing patch-wise contrastive constraints and Sinkhorn OT loss to realize deep manifold alignment via rigorous local and global distributional coherence. This comprehensive approach rectifies domain discrepancies, projecting the LR input into a sensor-invariant latent manifold $\mathbf{Z}_{\mathrm{LR}}$ that is manifold-aligned with the HR target $\mathbf{Z}_{\mathrm{HR}}$. This alignment step is specifically designed to reduce the initial source-target discrepancy, thereby facilitating a more regularized and less curved generative transport path.

Operating within this unified sensor-invariant latent space, the second phase, controllable residual generation, aims to enhance the efficiency and performance of generative SR. We propose the LSIRCM, incorporating a novel AnchorDiT backbone. As depicted in Fig. 4(b), this architecture

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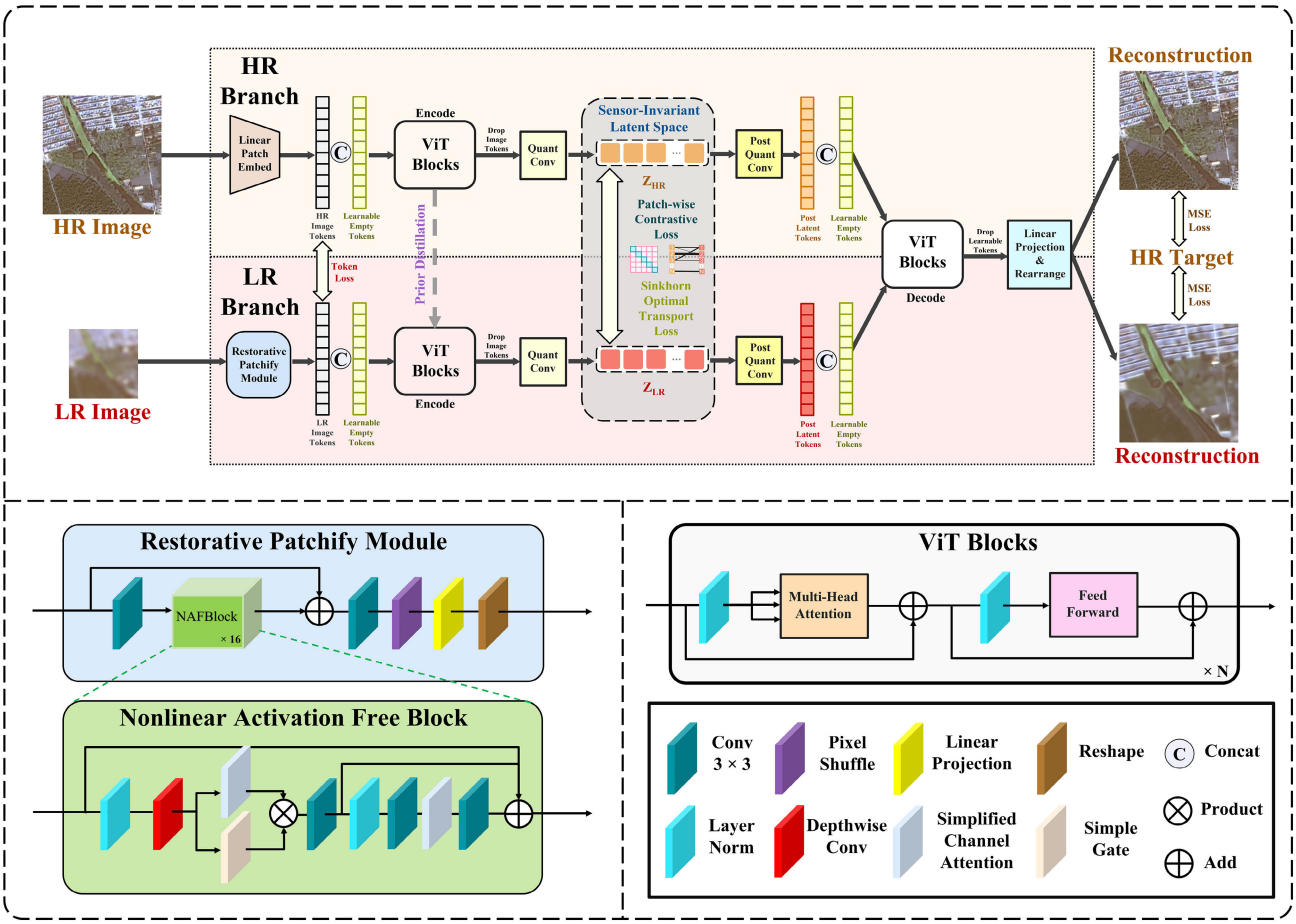


Fig. 5. Architecture of the proposed DSSC-MAE. The framework employs an asymmetric ViT-based design to construct a sensor-invariant latent space. (Bottom) Student branch incorporates an RPM to recover structural details from LR inputs, replacing the standard linear embedding used in (top) teacher branch. Learnable empty tokens are concatenated with image patches to compress spatial information into compact latent representations ($\mathbf{Z}_{\mathrm{HR}}$ and $\mathbf{Z}_{\mathrm{LR}}$). Hierarchical alignment is enforced via pixel/token fidelity, patch-wise contrastive loss, and sinkhorn OT.

reformulates SR as a latent residual generation task defined by the target $\Delta\mathbf{Z} = \mathbf{Z}_{\mathrm{HR}} - \mathbf{Z}_{\mathrm{LR}}$. Crucially, AnchorDiT utilizes the aligned latent priors $\mathbf{Z}_{\mathrm{LR}}$ as a structural anchor injected via cross-attention, enabling the model to map noisy states from any time step of the probability flow ODE trajectory directly to the clean residual. The model is optimized via a hybrid consistency training strategy that synergizes a self-target consistency objective $\mathcal{L}_{\mathrm{cons}}$ with a weighted reconstruction constraint $\mathcal{L}_{\mathrm{recon}}$. To balance generative diversity with structural precision, we devise a noise-aware weighting mechanism $w(\sigma_t)$ along the ODE trajectories. This strategy dynamically partitions the optimization process into a stochastic generation phase and a structural refinement phase, significantly enhancing both training stability and reconstruction fidelity in a single inference step.

C. Phase 1: Sensor-Invariant Manifold Learning via DSSC-MAE

Phase 1 introduces the DSSC-MAE to serve as a unified, domain-aligned image tokenizer. By adopting an asymmetric dual-stream architecture with a shared decoder, the framework implements a query-driven semantic compression mechanism. This design distills essential structural semantics into a compact latent manifold, filtering out sensor-specific redundancy to achieve highly efficient representation compression. Leveraging this architecture, a sequential optimization protocol is employed to explicitly bridge cross-sensor discrepancies. Stage I establishes a robust structural prior within the HR target domain. Subsequently, Stage II aligns the LR source domain to this established prior. This is achieved through a hierarchical alignment strategy that integrates pixel-level fidelity and token-level matching for spatial reconstruction, while synergizing patch-wise contrastive constraints and Sinkhorn OT to enforce rigorous local and global distributional coherence for deep manifold alignment. Consequently, the proposed DSSC-MAE constructs a high-quality sensor-invariant latent space, laying a solid foundation for the subsequent CM to achieve efficient single-step SR.

1) Architectural Specification: Dual-Stream Design With Query-Driven Semantic Compression: The proposed DSSC-MAE adopts an asymmetric architecture grounded on ViT backbones, as illustrated in Fig. 5. Given input batches with dimensions $I \in \mathbb{R}^{B\times C\times H\times W}$, the system incorporates a query-driven semantic compression mechanism to distill compact structural representations. This establishes a strictly constrained information bottleneck, compelling the model to encode structural semantics into a compact manifold while actively filtering out sensor-specific redundancy. By

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minimizing domain-dependent noise, this efficient abstraction lays a structural foundation that facilitates the subsequent hierarchical alignment strategies to explicitly bridge cross-sensor discrepancies.

In the teacher branch, HR reference images I_{HR} are initially processed using a standard linear patch embedding layer $\mathcal{P}(\cdot)$ to generate a sequence of N patch tokens. To facilitate robust manifold learning, particularly during the Stage I self-supervised pretraining, we incorporate a random masking strategy controlled by a masking ratio γ. Specifically, a subset of patch tokens is discarded, leaving only the visible tokens for processing. To execute semantic compression, we instantiate a set of M learnable latent tokens (also referred to as semantic queries) $\mathbf{T}_{\mathrm{empty}} \in \mathbb{R}^{M\times D}$. Here, M (where $M \ll N$) defines the dimensionality of the target latent manifold $\mathbf{Z}$. Unlike standard patch tokens bound to local spatial positions, these learnable tokens act as active semantic anchors, initialized independently of the input. These anchors are concatenated with the (potentially masked) patch tokens to form the composite input sequence

$$\mathbf{X}_{\mathrm{in}} = \left[\mathrm{Mask}\left(\mathcal{P}\left(I_{\mathrm{HR}}\right), \gamma\right) \,\|\, \mathbf{T}_{\mathrm{empty}}\right] \tag{1}$$

where $\|$ denotes the concatenation operation. This sequence is then processed by the ViT encoder layers, allowing $\mathbf{T}_{\mathrm{empty}}$ to aggregate global semantic information from the visible patches via self-attention mechanisms. Upon completion of encoding, a latent extraction operation discards the spatial image tokens, retaining only the updated query tokens to form the final representation

$$\mathbf{Z}_{\mathrm{HR}} = \mathrm{Extract}\left(\mathrm{ViT}\left(\mathbf{X}_{\mathrm{in}}\right)\right) = \mathbf{X}_{\mathrm{out}}\left[N : N+M\right]. \tag{2}$$

This strategy effectively decouples the semantic representation from the rigid pixel grid. By encouraging the network to aggregate information into a fixed set of tokens, the propagation of high-frequency sensor artifacts is minimized, thereby enhancing the robustness of the subsequent contrastive alignment.

Conversely, the student branch processes LR inputs I_{LR}. To mitigate the loss of high-frequency structural details typical of LR imagery, the standard linear projection is replaced with a specialized RPM. Constructed from a stack of nonlinear activation free blocks (NAFBlocks) [59] followed by a pixel-shuffle upsampler [60], the RPM acts explicitly as a structural information restorer. It is designed to regenerate geometric details and align the low-level feature distribution of the LR input with the teacher's domain prior before tokenization

$$F_{\mathrm{adapt}} = \mathrm{RPM}\left(I_{\mathrm{LR}}\right). \tag{3}$$

Similar to the teacher, these aligned features are concatenated with learnable queries. To realize the subsequent hierarchical alignment strategy, we employ a token-level distillation mechanism where intermediate features and output tokens of the student are constrained to match the frozen teacher. This ensures that the student's semantic abstraction hierarchy mirrors the teacher's high-fidelity prior.

Finally, the shared decoder utilizes a position-aware reconstruction mechanism to map the compressed latent manifold back to the pixel space. Compressed latent codes $\mathbf{Z}$ serve as semantic context, interacting with learnable positional queries to retrieve spatial details. The pipeline concludes with a patch-wise linear projection to assemble the reconstructed image I'_{rec}. Crucially, this decoding process serves as a supervisory signal to ensure that the learned manifold $\mathbf{Z}_{\mathrm{LR}}$ is geometrically aligned with $\mathbf{Z}_{\mathrm{HR}}$, maximizing the generative fidelity of the subsequent Phase 2.

2) Stage I: High-Fidelity Manifold Construction: The goal of this initial stage is to construct a well-structured and robust compressed latent space specifically for the HR domain. Following a self-supervised masked autoencoding paradigm, the teacher encoder E_{HR} and the shared decoder D are optimized to capture intrinsic image statistics. Given an input HR image I_{HR}, the encoder processes the partially masked sequence to derive a compact latent representation $\mathbf{Z}_{\mathrm{HR}}$ encapsulated within the learnable empty tokens

$$\mathbf{Z}_{\mathrm{HR}} = E_{\mathrm{HR}}\left(I_{\mathrm{HR}}, \gamma\right). \tag{4}$$

Subsequently, the shared decoder D reconstructs the original pixel-space signal from these compressed tokens

$$I'_{\mathrm{HR}} = D\left(\mathbf{Z}_{\mathrm{HR}}\right). \tag{5}$$

The training is driven by a composite objective comprising a primary reconstruction loss and a specialized auxiliary structural loss

$$\mathcal{L}_{\mathrm{Stage\ I}} = \mathcal{L}_{\mathrm{rec}}\left(I'_{\mathrm{HR}}, I_{\mathrm{HR}}\right) + \lambda_{\mathrm{aux}}\mathcal{L}_{\mathrm{aux}}. \tag{6}$$

The reconstruction term ensures perceptually faithful decoding by combining a pixel-wise ℓ_1 loss with an adversarial objective to mitigate oversmoothing and capture high-frequency textures. Following established generative reconstruction paradigms [29], the total reconstruction loss is formulated as:

$$\mathcal{L}_{\mathrm{rec}} = \|I_{\mathrm{HR}} - I'_{\mathrm{HR}}\|_1 + \lambda_{\mathrm{adv}}\mathcal{L}_{\mathrm{adv}} \tag{7}$$

where λ_{adv} balances pixel-wise consistency and perceptual realism. The adversarial component $\mathcal{L}_{\mathrm{adv}}$ leverages a PatchGAN-based MaskBit discriminator [61], employing a hinge loss formulation [62] to stabilize training dynamics and promote the synthesis of fine-grained structures.

To explicitly reinforce low-level geometric preservation, we incorporate an auxiliary histogram of oriented gradients (HOGs) loss [63], denoted as $\mathcal{L}_{\mathrm{aux}}$. This mechanism utilizes a lightweight auxiliary decoder D_{aux} (independent of the main image decoder) to predict geometric descriptors directly from the latent manifold. Let $G_{\mathrm{HOG}}(\cdot)$ represent a fixed, nonlearnable HOG feature extractor that computes the ground-truth gradient orientation maps $\mathbf{H}_{\mathrm{gt}} = G_{\mathrm{HOG}}(I_{\mathrm{HR}})$. The auxiliary decoder regresses these features from the latent codes, yielding $\mathbf{H}_{\mathrm{pred}} = D_{\mathrm{aux}}(\mathbf{Z}_{\mathrm{HR}})$. The auxiliary loss is formulated as the mean squared error (mse) between the predicted and ground-truth HOG features

$$\mathcal{L}_{\mathrm{aux}} = \|\mathbf{H}_{\mathrm{pred}} - \mathbf{H}_{\mathrm{gt}}\|_2^2. \tag{8}$$

By compelling the latent space to reconstruct purely geometric HOG descriptors, this auxiliary task prevents the encoder from overfitting to low-frequency information. Instead, it promotes the encoding of rich, orientation-sensitive structural primitives, which are essential for high-fidelity SR.

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3) Stage II: Hierarchical Cross-Sensor Alignment: Building upon the high-fidelity latent prior in Stage I, the second stage focuses on explicitly bridging the cross-sensor domain gap. Here, the parameters of the pretrained teacher encoder E_{HR} and the shared decoder D are frozen. This is crucial to preserve the structural integrity of the compact latent manifold constructed in Stage I, ensuring that the alignment process does not degrade the established manifold structure. Optimization is exclusively restricted to the student encoder E_{LR}. Equipped with the specialized RPM, the student is fine-tuned to distill structural knowledge from the frozen teacher. In contrast to the masked modeling in Stage I, the student processes full, unmasked LR inputs to maximize the exploitation of available structural cues. The alignment is enforced via a hierarchical alignment strategy that progressively addresses discrepancies through two distinct levels comprising traditional spatial reconstruction and deep manifold alignment.

To ensure fundamental spatial reconstruction, constraints are first imposed on pixel-level fidelity and token-level matching. The student's latent codes are decoded by the frozen decoder, enforcing pixel-level consistency between the reconstructed output I'_{SR} and the HR ground truth

$$\mathcal{L}_{\mathrm{SR}} = \|I'_{\mathrm{SR}} - I_{\mathrm{HR}}\|_1. \tag{9}$$

Simultaneously, to guide shallow feature extraction, a token-level feature alignment loss is introduced. This objective minimizes the mse between the teacher's linear embedding tokens, T_{HR}, and the student's RPM-refined tokens, T_{LR}

$$\mathcal{L}_{\mathrm{Token}} = \frac{1}{N}\sum_{i=1}^{N} \|T_{\mathrm{LR}}^{(i)} - T_{\mathrm{HR}}^{(i)}\|_2^2. \tag{10}$$

By aligning feature distributions immediately after the embedding stage, this constraint ensures that the student projects sensor-specific inputs into a semantic space compatible with the teacher's prior before deep encoding commences.

More critically, to rectify complex structural shifts and disentangle sensor-invariant content, a deep manifold alignment strategy is implemented. This is achieved by synergizing patch-wise contrastive constraints and Sinkhorn OT to enforce rigorous local and global distributional coherence.

We adopt a patch-wise contrastive learning objective, specifically PatchNCE [36], to facilitate local semantic alignment without paired pixel supervision in the feature domain. The underlying premise is that the latent representation of a specific spatial patch in the student domain should be maximally similar to its corresponding patch in the teacher domain (positive sample), while remaining distinct from other unrelated patches (negative samples). We formulate this objective as

$$\mathcal{L}_{\mathrm{NCE}} = -\sum_{i=1}^{L} \log \frac{\exp\left(\mathrm{sim}\left(z_{\mathrm{LR}}^{(i)}, z_{\mathrm{HR}}^{(i)}\right)/\tau\right)}{\sum_{j=1}^{L}\exp\left(\mathrm{sim}\left(z_{\mathrm{LR}}^{(i)}, z_{\mathrm{HR}}^{(j)}\right)/\tau\right)} \tag{11}$$

where $\mathrm{sim}(\cdot,\cdot)$ denotes cosine similarity and τ is the temperature parameter. Unlike standard image-wise contrastive learning, this patch-wise formulation preserves spatial specificity, ensuring the student captures fine-grained geometric primitives.

Complementing this local alignment, a Sinkhorn OT loss is incorporated to rectify global distributional shifts. This ensures that the overall density of the student's latent manifold is congruent with the teacher's manifold structure. The Sinkhorn loss minimizes the transport cost between the student distribution μ and teacher distribution ν

$$\mathcal{L}_{\mathrm{Sink}} = \sum_{i,j} \pi_{i,j}\mathbf{C}_{i,j} + \epsilon H(\pi) \tag{12}$$

where $\mathbf{C}_{i,j}$ represents the cosine distance matrix, $H(\pi)$ is the entropic regularization term, and π^* is the OT plan obtained via the Sinkhorn–Knopp algorithm [64]. This acts as a global regularizer that synergizes with the local PatchNCE to complete the deep manifold alignment.

The total optimization objective for Stage II combines these integrated constraints

$$\mathcal{L}_{\mathrm{Stage\ II}} = \lambda_{\mathrm{SR}}\mathcal{L}_{\mathrm{SR}} + \lambda_{\mathrm{Token}}\mathcal{L}_{\mathrm{Token}} + \lambda_{\mathrm{NCE}}\mathcal{L}_{\mathrm{NCE}} + \lambda_{\mathrm{Sink}}\mathcal{L}_{\mathrm{Sink}}. \tag{13}$$

This progressive strategy ensures that $\mathbf{Z}_{\mathrm{LR}}$ is rigorously aligned with the target $\mathbf{Z}_{\mathrm{HR}}$, thereby establishing a unified sensor-invariant manifold that serves as the robust structural basis for the subsequent generative phase.

D. Phase 2: Controllable Residual Generation via LSIRCM

Building upon the manifold-aligned sensor-invariant latent space established in Phase 1, the second phase addresses the critical tradeoff between generative fidelity and inference efficiency. To overcome the prohibitive inference latency and structural instability of traditional diffusion models in cross-sensor scenarios, we propose the LSIRCM. As illustrated in Fig. 4(b), this framework fundamentally reformulates SR as a structurally anchored latent residual generation task. To realize this, we employ the specialized AnchorDiT that integrates sensor-invariant priors directly into the generative process. The model is optimized through a hybrid consistency training strategy augmented by noise-aware weighting. This approach effectively mitigates solution drift by balancing trajectory consistency with fidelity constraints. Consequently, the network condenses the complex iterative prior into a deterministic mapping, enabling the prediction of clean residuals in a single forward pass without relying on pretrained teachers.

1) Reformulating SR as Latent Residual Generation via Consistency Training: To bridge the gap between high-fidelity reconstruction and real-time inference, we reformulate the SR task. Instead of learning a complex mapping from noise to the full HR distribution, we propose a latent residual consistency paradigm. Let $\mathbf{Z}_{\mathrm{HR}}$ and $\mathbf{Z}_{\mathrm{LR}}$ denote the latent codes extracted by the frozen encoders. We redefine the generative target as the structural residual $\Delta\mathbf{Z} = \mathbf{Z}_{\mathrm{HR}} - \mathbf{Z}_{\mathrm{LR}}$.

This reformulation provides two strategic advantages necessitated by efficient consistency training. First, it introduces structural anchoring. By leveraging the manifold-aligned $\mathbf{Z}_{\mathrm{LR}}$ from Phase 1 as a geometric foundation, the generative task is simplified from synthesizing a complete scene to recovering high-frequency textural details. Second, it yields trajectory simplification. The residual distribution is inherently zero-centered and sparse. Compared to the complex manifold of

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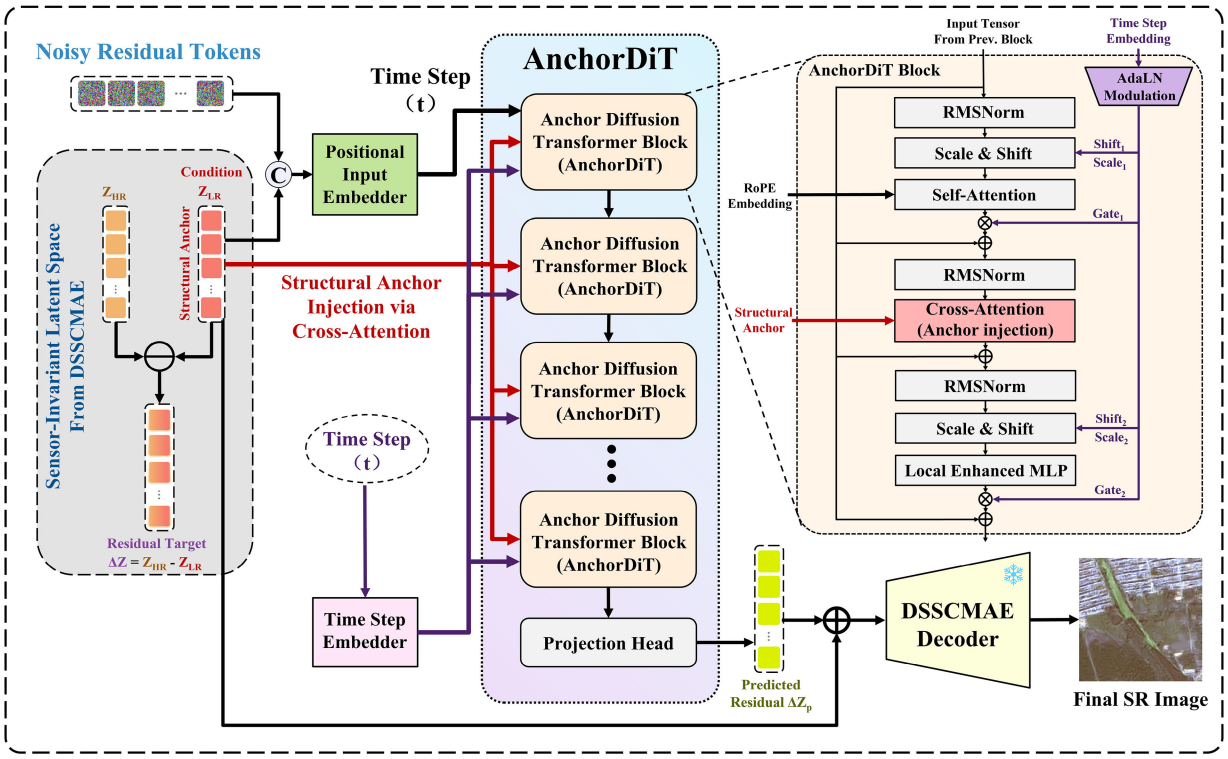


Fig. 6. Detailed network architecture of the LSIRCM. The core AnchorDiT backbone processes noisy residual tokens through a stack of blocks, where a cross-attention mechanism is employed to inject the sensor-invariant latent $\mathbf{Z}_{\mathrm{LR}}$ as a structural anchor. This design strictly conditions the generative process on robust geometric priors, guiding the progressive feature refinement to ensure high-fidelity residual reconstruction.

full latent distributions, the transport dynamics from noise to $\Delta\mathbf{Z}$ exhibit significantly reduced curvature, thereby facilitating the learning of a robust single-step consistency mapping.

However, given the reduced variance of the residual space, standard Gaussian noise schedules may overwhelm the sparse signal. To mitigate this, we introduce a noise scaling coefficient κ to explicitly modulate the perturbation magnitude. The adapted forward process is defined as

$$\mathbf{X}_t = \Delta\mathbf{Z} + \kappa\sigma_t\boldsymbol{\epsilon}, \quad \text{where } \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{14}$$

where σ_t represents the time-dependent noise level, equivalent to t in our continuous formulation, and κ balances the noise intensity against the energy of the residual features. Consequently, the core objective is to enforce the self-consistency property, requiring the model f_θ to map any noisy state $\mathbf{X}_t$ along the probability flow ODE trajectory directly to the clean residual origin

$$f_\theta(\mathbf{X}_t, t, \mathbf{Z}_{\mathrm{LR}}) = f_\theta(\mathbf{X}_r, r, \mathbf{Z}_{\mathrm{LR}}) \approx \Delta\mathbf{Z}, \quad \forall t, r \in [0, T]. \tag{15}$$

Realizing this mapping requires a neural backbone capable of processing the compressed semantic tokens produced by our tokenizer while effectively integrating the structural condition $\mathbf{Z}_{\mathrm{LR}}$.

2) LSIRCM Network Architecture: To accurately approximate the consistency mapping function f_θ defined above, we implement the LSIRCM using a specialized AnchorDiT backbone. The adoption of a Transformer-based architecture over conventional Convolutional U-Nets is strategically driven by the tokenized nature of the sensor-invariant latent space established in Phase 1. Since the query-driven compression produces a compact sequence of semantic tokens rather than dense spatial maps, Transformers are inherently adept at processing such representations. Crucially, U-Nets would require reshaping these tokens back into spatial grids, a process that risks introducing domain conversion artifacts and disrupting the learned manifold structure. AnchorDiT avoids this by operating directly within the tokenized space, preserving high-level semantic integrity.

As illustrated in Fig. 6, the AnchorDiT orchestrates a multistream inference pipeline designed to map the noisy residual state $\mathbf{X}_t$ directly to the clean prediction $\Delta\mathbf{Z}_{\mathrm{pred}}$. The architecture is structured to process three distinct modalities: the noisy residual tokens as the primary feature stream, the diffusion time step t as a global modulator, and the sensor-invariant latent $\mathbf{Z}_{\mathrm{LR}}$ as a structural condition. Features are scaled by noise level and timestep for numerical stability.

The pipeline commences by projecting the heterogeneous inputs of the primary stream and the modulator into a unified latent feature space. Specifically, the time step t is encoded into a global modulation vector c via a sinusoidal embedding followed by the time step embedder. Guided by the approach in [65], the sinusoidal embedding for time step t at dimensions $2k$ and $2k+1$ is formulated as

$$\begin{aligned} \mathrm{Sinusoidal}(t)_{2k} &= \sin\left(t/10000^{2k/D}\right) \\ \mathrm{Sinusoidal}(t)_{2k+1} &= \cos\left(t/10000^{2k/D}\right) \end{aligned} \tag{16}$$

where D is the embedding dimension. This continuous representation facilitates stable training. Simultaneously, the noisy residual tokens $\mathbf{X}_t$ are processed by a positional input embedder, where they are linearly projected and summed with

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learnable absolute position embeddings $\mathbf{P}_{\text{abs}}$ to form the initial hidden state $\mathbf{H}_0$. These initialization steps prepare the inputs for the subsequent transformer blocks and are formulated as

$$\begin{aligned} c &= \text{MLP}\left(\text{Sinusoidal}\left(t\right)\right) \\ \mathbf{H}_0 &= \text{Linear}\left(\mathbf{X}_t\right) + \mathbf{P}_{\text{abs}}. \end{aligned} \tag{17}$$

The core computation is performed by a stack of L AnchorDiT blocks, defined recursively as $\mathbf{H}_l = \mathcal{B}(\mathbf{H}_{l-1}, c, \mathbf{Z}_{\text{LR}})$. Within each block $\mathcal{B}$, the feature stream undergoes sequential refinement driven by three mechanisms. First, to incorporate global context, we utilize adaptive layer norm (AdaLN) [33] to modulate the features, followed by multihead self-attention (MHSA) equipped with rotary positional embeddings ($\mathcal{R}$) [66] to capture long-range dependencies. Let $\hat{\mathbf{H}} = (1 + \gamma_1(c)) \cdot \text{Norm}(\mathbf{H}_{l-1}) + \beta_1(c)$ denote the modulated input, where $\gamma_1(c)$, $\beta_1(c)$, and $\alpha_1(c)$ are the scale, shift, and gate parameters regressed from the global context c. The self-attention update is computed as

$$\begin{aligned} &\mathbf{Q} = \hat{\mathbf{H}}\mathbf{W}_Q, \quad \mathbf{K} = \hat{\mathbf{H}}\mathbf{W}_K, \quad \mathbf{V} = \hat{\mathbf{H}}\mathbf{W}_V \\ &\mathbf{H}'_l = \mathbf{H}_{l-1} + \alpha_1\left(c\right) \cdot \text{Softmax}\left(\frac{\mathcal{R}\left(\mathbf{Q}\right)\mathcal{R}\left(\mathbf{K}\right)^\top}{\sqrt{d_k}}\right)\mathbf{V}. \end{aligned} \tag{18}$$

Here, $\alpha_1(c)$ acts as an adaptive gating coefficient generated by the AdaLN module. Initialized to zero, it allows the block to approximate an identity mapping in the early training stages, thereby significantly stabilizing the optimization dynamics of deep architectures.

Subsequently, to enforce geometric consistency, the pipeline executes a structural anchor injection via cross-attention. To align the generation with the source modality, the feature stream enters a cross-attention interface. Unlike the self-attention mechanism, this module is conditioned on the aligned sensor-invariant latent $\mathbf{Z}_{\text{LR}}$, which acts as a fine-grained structural anchor. Here, the intermediate features $\mathbf{H}'_l$ provide the queries, while $\mathbf{Z}_{\text{LR}}$ supplies the keys and values

$$\begin{aligned} \mathbf{Q}_c &= \text{Norm}\left(\mathbf{H}'_l\right)\mathbf{W}_{Qc} \\ \mathbf{K}_c &= \mathbf{Z}_{\text{LR}}\mathbf{W}_{Kc}, \quad \mathbf{V}_c = \mathbf{Z}_{\text{LR}}\mathbf{W}_{Vc} \\ \mathbf{H}''_l &= \mathbf{H}'_l + \text{Softmax}\left(\frac{\mathbf{Q}_c\mathbf{K}_c^\top}{\sqrt{d_k}}\right)\mathbf{V}_c. \end{aligned} \tag{19}$$

By attending to $\mathbf{Z}_{\text{LR}}$ at the token level, this mechanism explicitly anchors the generated high-frequency residuals to the verified spatial layout of the LR input, effectively preventing geometric hallucinations.

The features are then refined by a locality-enhanced feed-forward network (LE-FFN). Addressing the lack of local inductive biases in standard Transformers, this module employs a SwiGLU activation structure augmented with a 3×3 depth-wise convolution (DWConv). Given the modulated input $\tilde{\mathbf{H}} = (1 + \gamma_2(c)) \cdot \text{Norm}(\mathbf{H}''_l) + \beta_2(c)$, the block output $\mathbf{H}_l$ is derived as

$$\begin{aligned} \mathbf{F} &= \text{SiLU}\left(\text{DWConv}\left(\tilde{\mathbf{H}}\mathbf{W}_1\right)\right) \odot \left(\tilde{\mathbf{H}}\mathbf{W}_2\right) \\ \mathbf{H}_l &= \mathbf{H}''_l + \alpha_2\left(c\right) \cdot \left(\mathbf{F}\mathbf{W}_3\right) \end{aligned} \tag{20}$$

where $\alpha_2(c)$ denotes the corresponding adaptive gating parameter for the FFN block. This mechanism ensures that the injection of nonlinear features is dynamically controlled by the diffusion time step, optimizing the residual consistency learning process.

Finally, after passing through L blocks, the deep features $\mathbf{H}_L$ are normalized and projected back to the residual space dimensions to yield the final prediction

$$\Delta\mathbf{Z}_{\text{pred}} = \text{Linear}_{\text{out}}\left(\text{Norm}\left(\mathbf{H}_L\right)\right). \tag{21}$$

This clean residual is then superimposed onto the LR latent $\mathbf{Z}_{\text{LR}}$ and passed through the shared decoder D to reconstruct the final super-resolved image.

3) Hybrid Consistency Training Strategy With Noise-Aware Weighting: Having defined the model architecture, the subsequent challenge lies in effectively optimizing the AnchorDiT parameters to approximate the consistency mapping function f_θ. We implement a hybrid consistency training strategy tailored to address the challenge of processing heterogeneous sensor data with complex noise distributions. By integrating a trajectory consistency objective with a noise-modulated fidelity constraint, this strategy ensures that the model simultaneously learns the global transport dynamics and preserves local structural accuracy.

The training process initiates by adding noise to the target residuals, sampling continuous time steps t according to a log-normal schedule. This sampling strategy is selected to bias the optimization toward noise levels relevant to perceptual quality, ensuring that the model focuses on restoring critical structural frequencies rather than fitting high-variance Gaussian noise. To implement this, the total optimization objective is formulated as a weighted sum of two complementary terms

$$\mathcal{L}_{\text{total}} = \lambda_{\text{cons}}\mathcal{L}_{\text{cons}} + \mathcal{L}_{\text{recon}}. \tag{22}$$

The primary component, the consistency loss $\mathcal{L}_{\text{cons}}$, enforces the self-consistency property of the probability flow ODE. Departing from distillation methods that depend on pretrained teachers, we adopt a self-training paradigm to learn the generative prior directly from scratch. In this scheme, the regression target is generated by the online network f_θ itself via a stop-gradient operator, effectively stabilizing the moving-target optimization. Mathematically, this loss minimizes the distance between the current predicted residual and a target derived from a single step of the ODE solver toward a subsequent time r

$$\mathcal{L}_{\text{cons}} = \mathbb{E}_{t,\mathbf{X}}\left[\frac{\left\|f_\theta\left(\mathbf{X}_t, t, \mathbf{Z}_{\text{LR}}\right) - \text{sg}\left(f_\theta\left(\mathbf{X}_r, r, \mathbf{Z}_{\text{LR}}\right)\right)\right\|_2^2}{|t - r| + \epsilon}\right] \tag{23}$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operation, and the denominator normalizes the gradient magnitude across varying time intervals. For log-sampled timesteps, the subsequent step r ($r < t$) is dynamically derived from t using a stage-aware decay function to stabilize ODE trajectory learning

$$r = \max\left(t \cdot \left(1 - \frac{1 + k \cdot \sigma\left(-bt\right)}{q^{s+1}}\right), 0\right) \tag{24}$$

where $\sigma(\cdot)$ is the sigmoid function; s is the current training stage; and q, k, and b are hyperparameters. This formulation adaptively adjusts the interval $|t - r|$ across different magnitudes of t and progressively shrinks it as training advances, stabilizing the consistency mapping.

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Complementing the consistency objective, the reconstruction loss $\mathcal{L}_{\text{recon}}$ serves as a strictly grounded fidelity constraint. While consistency ensures valid mapping paths, unconstrained generation in cross-sensor scenarios often suffers from solution drift. This term functions as a weighted regression objective, enforcing the generated residual to converge to the ground truth $\Delta\mathbf{Z}$ within specific noise regimes. The impact of this loss is governed by a noise-aware weighting function $w(\sigma_t)$

$$\mathcal{L}_{\text{recon}} = \mathbb{E}_{t,\mathbf{X}}\left[w(\sigma_t)\cdot\|f_\theta(\mathbf{X}_t,t,\mathbf{Z}_{\text{LR}})-\Delta\mathbf{Z}\|_2^2\right]. \tag{25}$$

To dynamically balance generative diversity with structural precision, we model the weighting function $w(\sigma_t)$ as a soft gating mechanism, mathematically equivalent to a high-order Butterworth low-pass filter

$$w(\sigma_t) = \frac{1}{1+\left(\frac{\sigma_t}{\sigma_{\text{fidelity}}}\right)^k} \tag{26}$$

where σ_{fidelity} controls the cutoff noise level and k determines the transition steepness. This mechanism effectively partitions the optimization landscape along the ODE trajectory into two distinct regimes. In the structural refinement regime (where $\sigma_t < \sigma_{\text{fidelity}}$), the weight $w(\sigma_t)$ approaches unity, strictly enforcing alignment with the pure target to prevent spectral shifts and geometric distortions. Conversely, in the stochastic generative regime (where $\sigma_t > \sigma_{\text{fidelity}}$), the weight decays to zero. This relaxation prioritizes the consistency objective, allowing the model to focus on learning global transport directions without being penalized for the inherent uncertainty of high-noise states.

4) Efficient Single-Step Inference: The culminating advantage of the proposed training paradigm is the capability to decouple high-fidelity generation from iterative refinement. During the inference phase, the LSIRCM framework effectively condenses the complex diffusion trajectory into a single deterministic mapping step. The procedure commences with extracting the sensor-invariant latent anchor $\mathbf{Z}_{\text{LR}}$ from the input LR image via the pretrained encoder. Subsequently, a standard Gaussian noise vector $\mathbf{X}_T$ is sampled, and the trained AnchorDiT backbone predicts the clean structural residual $\Delta\mathbf{Z}_{\text{pred}}$ in a single forward pass, strictly conditioned on $\mathbf{Z}_{\text{LR}}$.

The final super-resolved output is reconstructed by performing an additive integration of the predicted residual and the latent anchor, followed by decoding the fused representation:

$$I_{\text{SR}} = \text{Decoder}\left(\mathbf{Z}_{\text{LR}} + f_\theta(\mathbf{X}_T, T, \mathbf{Z}_{\text{LR}})\right). \tag{27}$$

This protocol fundamentally reduces the inference complexity to $\mathcal{O}(1)$, standing in stark contrast to the $\mathcal{O}(N)$ complexity of traditional diffusion models, where N typically spans hundreds to thousands of steps. Consequently, our method achieves a significant improvement in computational efficiency, making it suitable for time-sensitive, large-scale remote sensing monitoring tasks.

IV. Experiments

A. Datasets

To evaluate the robustness and scalability of the proposed framework under different sensor conditions, we conducted experiments on two distinct cross-sensor datasets: the public benchmark SEN2NAIP and a large-scale custom dataset, GF6-8x. The statistical specifications and land cover characteristics are provided in Table I. Visual examples illustrating the resolution and spectral differences between the LR inputs and HR ground truths for both datasets are shown in Fig. 7.

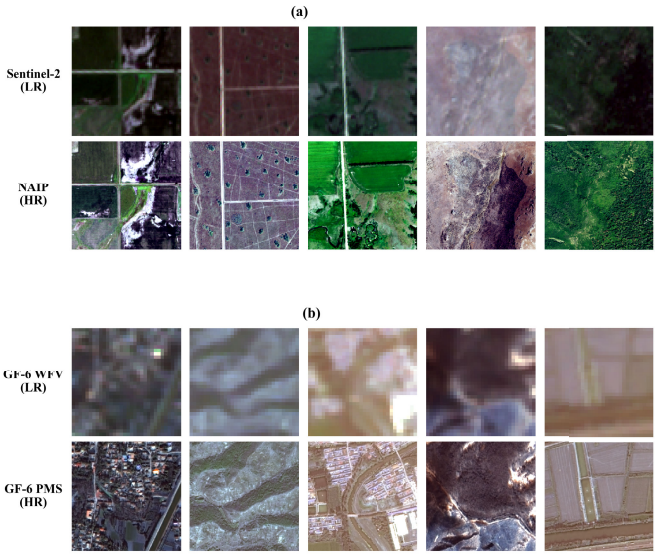


Fig. 7. Sample pairs from the cross-sensor datasets used for evaluation. (a) SEN2NAIP examples showing Sentinel-2 (LR) inputs and NAIP (HR) aerial ground truths. (b) GF6-8x examples showing GF-6 WFV (LR) inputs and GF-6 PMS (HR) satellite ground truths. All images are displayed as true-color composites (RGB). To improve visual clarity and normalize the radiometric ranges, a consistent cumulative histogram stretching (2%–98%) was applied to each pair.

TABLE I
Statistical Summary of Datasets Used in Experiments

Attribute	SEN2NAIP	GF6-8x
Task	Sentinel-2 → NAIP	GF-6 WFV → PMS
Scale Factor	4×	8×
Input Size (LR)	64 × 64	32 × 32
Output Size (HR)	256 × 256	256 × 256
Sensor Type	Multispectral (4 Bands)	Multispectral (4 Bands)
Land Cover	Predominantly Cropland	Diverse (Urban, Water, Forest, etc.)
Total Pairs	11,404	100,000
Train / Test	10,260 / 1,144	90,000 / 10,000
Alignment RMSE (px)	0.18	0.28

Note: The SEN2NAIP dataset consists of standardized patches generated via sliding-window cropping from the 2,851 original pairs in the official cross-sensor subset [22].

For the SEN2NAIP dataset (4× SR), we specifically utilized the cross-sensor subset defined in [22]. Unlike simulated datasets based on synthetic degradation, this benchmark subset originally consists of 2851 real-world, coregistered pairs of Sentinel-2 Level-2A satellite imagery (LR) and NAIP aerial photography (HR). This allows us to test the model against actual radiometric differences and spectral response variations between satellite and aerial sensors. Regarding scene composition, although the original images cover various regions in USA, the dataset is dominated by agricultural landscapes and semi-natural terrains, with limited coverage of dense urban areas. To standardize the spatial dimensions for deep learning training, we applied a sliding-window cropping strategy to these original image pairs, subdividing them into 11 404 standardized patches (256 × 256 HR/64 × 64 LR). These were subsequently split into training and testing sets with a 9:1 ratio.

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To address the scarcity of large-scale benchmarks for high-magnification multispectral SR, we constructed the custom GF6-8x dataset (8× SR) using imagery from the Gaofen-6 (GF-6) satellite. This dataset presents significant challenges due to the large scaling factor and high scene diversity. We leveraged the dual-sensor configuration of GF-6, which hosts both a wide field of view (WFV) sensor (16-m resolution) and an HR PMS sensor (2-m resolution) on a single platform. This configuration enables quasi-synchronous acquisition, which essentially minimizes temporal discrepancies and maintains consistent observation geometry. To improve geometric reliability, the GF-6 images were derived from radiometrically calibrated Level-4 products. The processing pipeline subsequently performed precise orthorectification and registration utilizing high-precision ground control points (GCPs) and digital elevation models (DEMs). In contrast to the agriculture-focused SEN2NAIP, the GF6-8x benchmark covers a diverse range of land cover categories, including water bodies, dense urban settlements, croplands, and forests. This semantic diversity encourages the model to generalize across complex spectral–spatial patterns. Following data preprocessing and cropping, we assembled a large-scale dataset of 100 000 paired samples with a 9:1 training–testing split.

Furthermore, we quantitatively evaluated the residual spatial misregistration using the structural phase correlation coefficient (PCC) in the native LR domain. As detailed in Table I, the GF6-8x dataset exhibits a subpixel alignment root-mean-square error (RMSE) of 0.28 pixels. This subpixel accuracy effectively validates the registration reliability of the dataset, providing a robust geometric foundation for cross-sensor SR training.

B. Implementation Details

The proposed framework was implemented in PyTorch. All experiments were executed on a single NVIDIA A100 GPU with a batch size of 16. Optimization was performed using the AdamW optimizer with an initial learning rate of 1×10^{-4} ($\beta_1 = 0.9, \beta_2 = 0.995$, and weight decay 0.005), modulated by a cosine annealing schedule.

In the first phase, the DSSC-MAE (utilizing a ViT-Base backbone with patch size 16) followed a two-stage optimization protocol. The pretraining stage spanned 250k iterations with a dynamic masking strategy: the mask ratio started at 0.25, linearly annealed to 0 between 200 and 230k iterations, and remained 0 thereafter. This schedule prioritizes structural extraction before full-context reconstruction. Subsequently, the model was fine-tuned for an additional 150k iterations to enforce sensor alignment. The loss coefficients were set to $\lambda_{\text{SR}} = 1.0$, $\lambda_{\text{token}} = 1.0$, $\lambda_{\text{NCE}} = 0.05$, and $\lambda_{\text{sink}} = 0.005$.

In the second phase, the parameters of the pretrained DSSC-MAE were frozen, and the LSIRCM was optimized for 300k iterations. The consistency loss weight was set to $\lambda_{\text{cons}} = 0.01$. To stabilize training, a noise-aware weighting strategy was applied with a data variance of $\sigma_{\text{data}} = 0.5$ and a decay factor of $\gamma = 12$. This design allows for efficient generative modeling within the compact latent space, avoiding the computational overhead of direct pixel-space synthesis.

For fair comparison, official open-source implementations were utilized for all baselines, strictly adhering to their original hyperparameter configurations. To ensure consistent data distributions, all models were retrained from scratch on the specific training splits of the SEN2NAIP and GF6-8x datasets until convergence.

C. Evaluation Metrics

To comprehensively assess reconstruction quality, both fidelity-oriented and perception-oriented metrics are employed.

1) Fidelity-Oriented Metrics: For pixel-level fidelity, standard community benchmarks are adopted: PSNR and structural similarity index (SSIM). PSNR quantifies the ratio of peak signal-to-noise power based on mse, while SSIM evaluates perceptual degradation by analyzing luminance, contrast, and structural information. Although widely used, these metrics inherently favor models optimized for mse, often resulting in smoothed high-frequency textures. This limitation highlights the perception–distortion tradeoff [8], necessitating the inclusion of perceptual metrics.

2) Perception-Oriented Metrics: To evaluate perceptual realism beyond pixel alignment, three advanced metrics are incorporated. Learned perceptual image patch similarity (LPIPS) [39] calculates distances in deep feature space, aligning closely with human visual perception. Deep image structure and texture similarity (DISTS) [67] focuses specifically on texture and structural details, penalizing the oversmoothing artifacts common in regression-based methods. Additionally, Fréchet inception distance (FID) [15] evaluates generation quality at the distributional level by measuring the divergence between generated and real data features. Collectively, lower scores for LPIPS, DISTS, and FID indicate superior perceptual quality.

D. Performance Evaluation

To evaluate the effectiveness of the proposed framework, we benchmark LSIRCM against a wide range of state-of-the-art methods. These include regression-based networks that prioritize signal fidelity (FSRCNN [68], EDSR [40], SwinIR [11], and HAT [41]) and generative models designed for perceptual recovery (MA-GAN [44], SR3 [17], EDiffSR [19], FastDiffSR [46], and LDM [27]). We conduct experiments on both the standard SEN2NAIP task and the more challenging GF6-8x task to verify performance across different sensor types and scaling factors.

1) Quantitative Comparison: Table II presents the quantitative evaluation on the SEN2NAIP (4×) and GF6-8x (8×) datasets. These results confirm that LSIRCM successfully balances the conflicting objectives of structural fidelity, perceptual realism, and inference speed, demonstrating a comprehensive performance advantage over existing cross-sensor methods.

On the standard SEN2NAIP dataset, baseline methods exhibit a clear performance tradeoff. While lightweight networks like FSRCNN lack the capacity to resolve complex domain gaps (21.53-dB PSNR and 155.54 FID), heavier regression-based approaches, such as SwinIR, prioritize pixel-level accuracy, yielding high scores in PSNR (22.75 dB)

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TABLE II

QUANTITATIVE COMPARISON ON SEN2NAIP (4×) AND GF6-8X (8×) DATASETS. STEPS DENOTE THE NUMBER OF INFERENCE SAMPLING STEPS. THE SYMBOLS ↑ AND ↓ INDICATE THAT HIGHER AND LOWER VALUES ARE BETTER, RESPECTIVELY. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN **RED** AND **BLUE**, RESPECTIVELY

Category	Method	Steps	SEN2NAIP (4×)					GF6-8x (8×)				
			PSNR↑	SSIM↑	LPIPS↓	DISTS↓	FID↓	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	FID↓
Regression -based	FSRCNN [68]	50	21.53	0.6464	0.5557	0.3096	155.54	28.24	0.7537	0.6111	0.3975	243.00
	EDSR [40]	1	22.51	0.6747	0.4674	0.2720	98.44	30.73	0.8026	0.3916	0.3030	125.39
	SwinIR [11]	1	**22.75**	**0.6808**	0.5029	0.2833	96.78	**30.88**	**0.8032**	0.4388	0.3236	116.33
	HAT [41]	1	**22.68**	**0.6782**	0.5258	0.2911	103.04	**30.88**	**0.8037**	0.4309	0.3199	116.33
Generative -based	MA-GAN [44]	1	22.41	0.6348	0.4140	0.2617	104.95	27.99	0.7087	0.4510	0.3252	155.47
	SR3 [17]	2000	18.84	0.5968	0.3839	0.2672	64.10	18.06	0.5754	0.3751	0.2589	72.04
	EDiffSR [19]	100	20.04	0.5567	**0.2959**	**0.2101**	**57.35**	27.49	0.6171	0.3029	0.2247	62.86
	LDM [27]	50	14.94	0.5477	0.4041	0.2510	60.81	20.57	0.6632	0.2964	0.2230	**39.37**
	FastDiffSR [46]	20	20.79	0.6399	0.3622	0.2484	62.05	24.86	0.7321	**0.2806**	**0.2155**	52.57
	LSIRCM (Ours)	1	22.37	0.6485	**0.2603**	**0.2018**	**56.10**	29.72	0.7685	**0.2140**	**0.1886**	**30.98**

and SSIM (0.6808). However, this optimization strategy leads to oversmoothed textures, resulting in poor perceptual performance with an LPIPS of 0.5029. In contrast, LSIRCM achieves superior perceptual quality across all metrics, reducing the LPIPS to 0.2603 and recording the best DISTS of 0.2018 and FID of 56.10. Critically, unlike standard diffusion models that often compromise structural integrity for texture, such as SR3 that drops to 18.84 dB in PSNR, and LDM that sharply drops to 14.94 dB, LSIRCM maintains a competitive PSNR of 22.37 dB and SSIM of 0.6485. This indicates that our model effectively overcomes the perception–distortion tradeoff, preserving structural similarity while synthesizing realistic details.

The robustness of our framework is even more pronounced on the challenging GF6-8x dataset. Unlike SEN2NAIP, this task involves a larger 8× scaling factor and distinct spectral shifts between GF-6 WFV and PMS sensors. In this regime, early CNN baselines experience severe performance degradation (e.g., FSRCNN yielding an FID of 243.00), and standard generative models face significant challenges due to latent misalignment. Specifically, SR3 suffers from severe structural degradation, with PSNR dropping to 18.06 dB and SSIM to 0.5754. LDM exhibits a comparable limitation, yielding a competitive FID of 39.37 but suffering from severely degraded fidelity metrics. This outcome substantiates our theoretical premise that operating within an unaligned latent space inherently complicates the generative flow. Consequently, the model becomes biased toward synthesizing plausible textures rather than reconstructing faithful structural details. Conversely, LSIRCM dominates the generative category across the entire metric suite. It achieves the lowest perceptual scores with an LPIPS of 0.2140, DISTS of 0.1886, and FID of 30.98, while recovering remarkable structural fidelity with a PSNR of 29.72 dB and SSIM of 0.7685. This represents an improvement of 11.66 dB over SR3 and 4.86 dB over FastDiffSR. It is noteworthy that the absolute metrics on GF6-8x are generally higher than on SEN2NAIP. This difference stems from intrinsic data characteristics, as the aerial NAIP imagery exhibits greater spatial variance and mean intensity than the satellite-based PMS data, inherently amplifying reconstruction errors on SEN2NAIP.

This consistent superiority validates the synergy of our proposed modules. The high PSNR and SSIM scores are directly attributed to the DSSC-MAE, which employs a hierarchical alignment strategy to explicitly align heterogeneous sensor representations within the latent space. Building on this robust structural foundation, the AnchorDiT architecture injects these aligned priors as anchors to enforce controllable generation, ensuring that the synthesized details exhibit high perceptual quality (low LPIPS, DISTS, and FID) and rigorous fidelity to the ground truth.

Importantly, we achieve this state-of-the-art performance with superior operational efficiency. Unlike diffusion baselines, including SR3 (2000 steps), EDiffSR (100 steps), LDM (50 steps), and FastDiffSR (20 steps) that require iterative sampling, LSIRCM operates via a deterministic single inference step. This demonstrates that our framework effectively combines the high-speed inference of regression networks with the generative quality of diffusion models. Consequently, LSIRCM establishes a new benchmark, achieving the best perceptual quality among all methods while simultaneously outperforming existing diffusion models in reconstruction fidelity.

2) Qualitative Comparison: While the numerical evaluations in Section IV-D1 indicate the performance advantage of LSIRCM, quantitative metrics alone cannot fully capture the perceptual fidelity required for expert interpretation in remote sensing. To rigorously assess the operational utility of our framework, we conduct a detailed visual examination of representative samples. For consistent analysis, all images are displayed using a standard false color composite where the near-infrared (NIR), red, and green bands are mapped to the R, G, and B channels, respectively. Additionally, consistent absolute linear stretching is applied to ensure that identical band values are rendered with the same color intensity across all comparisons. These visual results corroborate the quantitative findings, highlighting how different modeling approaches handle complex cross-sensor discrepancies.

On the SEN2NAIP dataset (4× scale, Fig. 8), distinct performance gaps are observable across different paradigms. Regression-based approaches, including the lightweight FSRCNN and heavier models like EDSR, SwinIR, and HAT, exhibit characteristic oversmoothing. While they preserve

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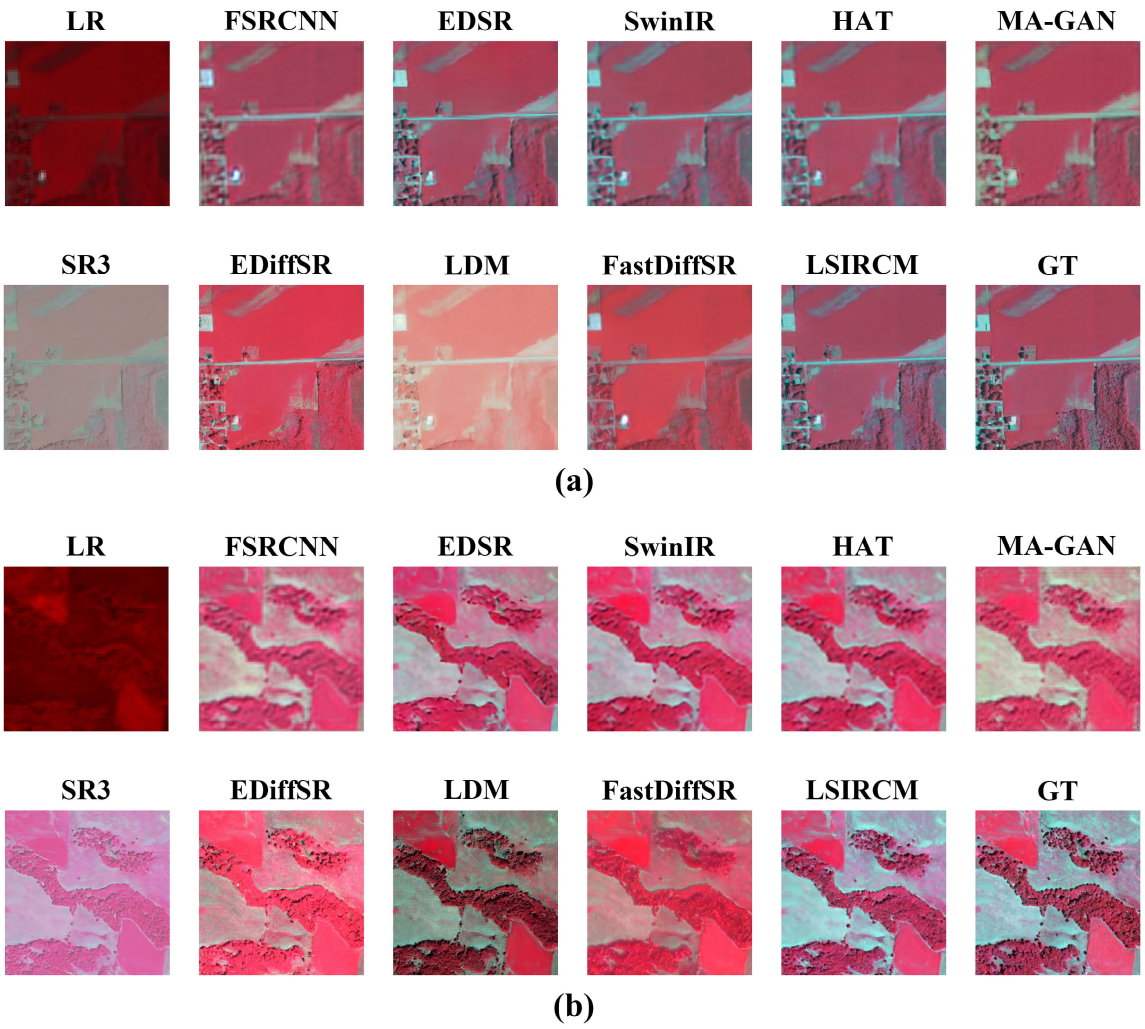


Fig. 8. Visual comparison on the SEN2NAIP dataset (4×) using standard false color composites (NIR-R-G → RGB). (a) Mixed land-cover scene comprising farmland, roads, and buildings. (b) Complex vegetation community scene. The proposed LSIRCM demonstrates superior spectral fidelity, exhibiting the minimal color shift compared to the ground truth. It effectively reconstructs authentic vegetation textures without introducing the spectral distortions or artifacts commonly observed in competing methods.

the global layout of the agricultural parcels, they fail to resolve high-frequency details, resulting in indistinct boundaries between diverse land cover types and a notable absence of fine-grained internal textures within vegetation areas. This observation aligns with their high PSNR but poor perceptual scores. In contrast, the GAN-based baseline suffers from visible artifacts and fails to maintain spatial continuity. Critically, standard diffusion models such as SR3, LDM, EDiffSR, and FastDiffSR exhibit limitations in cross-sensor scenarios. As shown in Fig. 8, both SR3 and LDM exhibit significant spectral distortion. They both present a faded, pale visual effect and appear either too bright or too dark, which differs significantly from the actual radiometric characteristics of ground truth. Conversely, our proposed LSIRCM demonstrates superior spectral fidelity, exhibiting the minimal spectral shift among all methods. It accurately reproduces the deep red hues indicative of healthy vegetation while precisely recovering authentic canopy structures in the forest area. This validates that our sensor-invariant manifold learning effectively bridges the sensor gap, preventing the color shifting and textural hallucinations common in unconstrained generative models.

The limitations of existing baselines become even more pronounced in the challenging 8× upscaling scenario of the GF6-8x dataset (Fig. 9). The lightweight baseline model FSRCNN produces overly blurry results, while heavier regression-based methods fail to recover the structural layout. In addition to reducing complex facilities to indistinguishable blurred patches, they exhibit excessive smoothing in vegetated regions, failing to capture essential high-frequency information. Severely impacted by the significant cross-sensor domain discrepancy, the GAN-based MA-GAN suffers from mode collapse, leading to a failure in maintaining structural coherence and resulting in severe structural distortion. Meanwhile, standard generative models such as SR3 and LDM suffer from severe spectral distortion. SR3 exhibits excessively low contrast and appears either too bright or too dark, while LDM displays an unnatural greenish-dark tint, both inconsistent with the radiometric characteristics of ground truth. EDiffSR, while yielding sharper edges, tends to generate artificial high-frequency noise. This artifact is particularly evident in homogeneous regions such as agricultural parcels, where it replaces smooth surfaces with grainy noise patterns.

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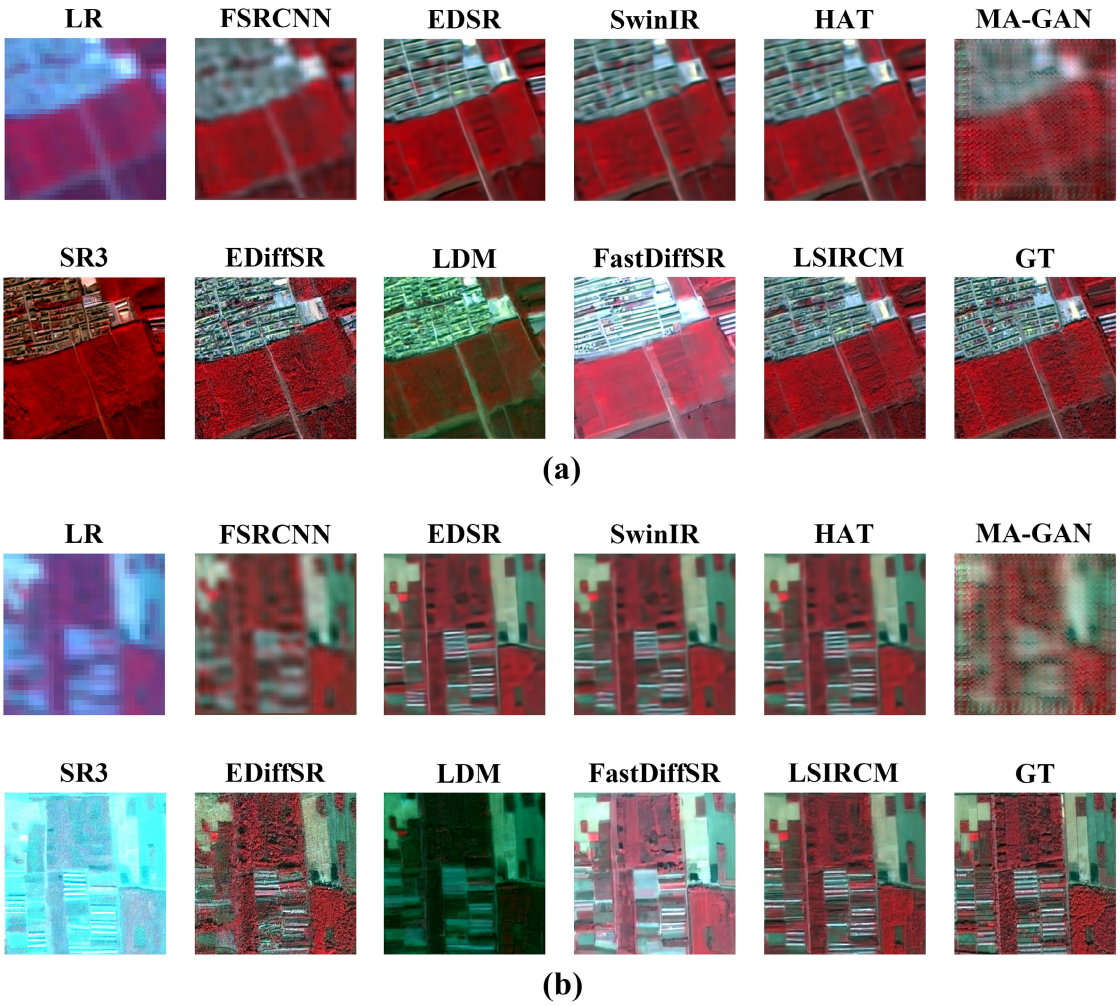


Fig. 9. Visual comparison on the GF6-8x dataset (8×) using standard false color composites (NIR-R-G → RGB). (a) Mixed urban and farmland scene. (b) Agricultural scene featuring farmlands and greenhouses. In this challenging large-scale factor scenario, LSIRCM excels in preserving fine-grained geometric structures and recovering realistic details. It avoids the structural hallucinations typical of generative models, maintaining high fidelity to the true ground features.

FastDiffSR, conversely, exhibits an unnatural structural inconsistency: it oversmooths internal fine details while excessively sharpening edges, resulting in the generation of numerous inaccurate structures that lack physical plausibility. Collectively, this tendency of diffusion-based models to generate false details fundamentally contradicts the core principle of faithful Earth observation. In contrast, LSIRCM effectively resolves these ambiguities. It precisely reconstructs the linear geometries of the agricultural facilities and recovers the natural texture of the dense vegetation. By strictly anchoring the generation to the aligned structural prior, our method avoids both the spectral deviations of SR3 and LDM and the textural artifacts of EDiffSR, maintaining high fidelity to the ground truth at this larger scale factor.

Distinct from these baselines, our proposed framework achieves an optimal balance between generative richness and structural fidelity. By strictly anchoring the generative process to aligned structural priors, we effectively mitigate sensor-specific variations and suppress hallucinations. This enhanced visual quality significantly improves interpretability for human experts, providing the structural accuracy or radiometric consistency required for reliable analysis. Notably, these perceptual gains are realized through efficient single-step inference, offering a decisive speed advantage over iterative diffusion models. Consequently, by unifying high-fidelity reconstruction with rapid processing capability, our framework establishes itself as a robust and practical solution for real-world cross-sensor remote sensing SR.

E. Ablation Study

To evaluate the contribution of individual components within the proposed framework, ablation experiments were conducted on the challenging GF6-8x dataset. We prioritized this dataset for its substantial volume of satellite-based imagery and high fidelity to operational scenarios, which capture inherent cross-sensor discrepancies across diverse land cover categories. These characteristics provide a rigorous testbed for assessing model robustness. To ensure a fair comparison, all variants were trained according to the protocols detailed in Section IV-B.

1) Evaluation of Sensor-Invariant Manifold Learning: To validate the efficacy of the sensor-invariant manifold learning in Phase 1, we performed a comprehensive ablation study

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TABLE III
EVALUATION OF BACKBONE ARCHITECTURES FOR MANIFOLD REPRESENTATION. MAE OUTPERFORMS KL-VAE IN BOTH RECONSTRUCTION FIDELITY AND LATENT MANIFOLD STRUCTURE (LOWER KL DIVERGENCE)

Method	PSNR ↑	SSIM ↑	FID ↓	KL ↓
KL-VAE	38.21	0.9520	12.01	0.2579
MAE (Baseline)	**42.38**	**0.9828**	**2.26**	**0.1990**

on the tokenizer architecture and the proposed hierarchical alignment strategy. We assessed the superiority of the MAE backbone over conventional latent mapping approaches and explicitly quantified the contributions of the RPM and the deep manifold alignment mechanism in establishing a strictly aligned latent space.

We first justify the selection of MAE as the backbone for manifold construction over the standard KL-VAE. As detailed in Table III, while KL-VAE serves as the default tokenizer for most LDMs, its probabilistic constraint forces the latent distribution toward a standard Gaussian. This regularization inevitably erodes the subtle high-frequency details essential for remote sensing interpretation, resulting in a constrained reconstruction PSNR of 38.21 dB. In contrast, the continuous representation of the MAE achieves significantly superior fidelity (42.38 dB) and establishes a semantically structured latent manifold, evidenced by a significantly lower KL divergence of 0.1990. This confirms that the MAE provides a higher fidelity continuous canvas, serving as a robust foundation for the subsequent restoration task.

Having established the backbone, we further conduct a fine-grained ablation to deconstruct the contributions of the RPM and the hierarchical alignment strategy. The quantitative results are consolidated in Table IV. The Stage I MAE baseline, trained solely on HR reconstruction without cross-sensor constraints, exhibits a substantial domain gap ($\ell_2 \approx 221.62$) and poor inference performance (PSNR 26.86 dB), highlighting the severity of the intrinsic sensor discrepancy. Furthermore, separating the deep alignment objectives demonstrates that ablating either the global Sinkhorn OT or the local patch-wise NCE results in increased structural divergence and perceptual degradation compared to the full framework.

Subsequently, we investigate the specific contribution of the RPM. Replacing the RPM with a standard linear embedding layer ("w/o RPM") results in a perceptible decline in reconstruction fidelity, with PSNR dropping from 30.81 to 30.65 dB. This result validates that the RPM functions as a critical structural information restorer, effectively recovering the missing high-frequency spatial details typical of LR imagery. Conversely, isolating the deep manifold alignment reveals the limitations of purely spatial supervision ($\mathcal{L}_{\text{SR}} + \mathcal{L}_{\text{Token}}$). While the "w/o Deep Manifold Alignment" variant yields the highest PSNR (30.85 dB) via aggressive pixel-wise fitting, it suffers from poor perceptual quality (LPIPS 0.4332) and significant structural divergence (KL 0.1092). This indicates that relying solely on traditional spatial supervision forces pixel-level regression without correcting the underlying manifold topology. Comparing these variants confirms that while the RPM enhances spatial reconstruction, integrating the Sinkhorn and PatchNCE constraints is the primary driver for manifold correction. By enforcing semantic consistency, this deep alignment significantly reduces structural divergence (KL drops to 0.0589) and residual magnitude (ℓ_2 drops to 155.40). Although this strictly constrained manifold incurs a marginal cost in PSNR (0.04 dB), it yields substantial gains in perceptual fidelity (LPIPS improves to 0.3585, FID to 99.24). This confirms that our full hierarchical strategy effectively harmonizes the heterogeneous distributions into a shared, structure-preserving latent space.

To visually corroborate these quantitative findings, we present the principal component analysis (PCA) visualization of the latent residual vectors ($\Delta\mathbf{z} = \mathbf{z}_{\text{HR}} - \mathbf{z}_{\text{LR}}$) in Fig. 10. To interpret the semantic distribution, data points are color-coded by clusters derived via CLIP feature extraction [69] and K-means clustering [70], with kernel density estimation (KDE) contours overlaid to visualize distributional concentration.

The Stage I baseline exhibits a high-variance dispersion with residuals scattered chaotically, indicating significant sensor misalignment. When decomposing the deep manifold alignment, the variant without patch-wise NCE retains only the OT constraint, which focuses on global alignment and effectively compresses the overall residual magnitude. Conversely, the variant w/o Sinkhorn OT loses this global constraint. Its patch-wise NCE strictly focuses on local semantic alignment, which produces richer structural separation but results in a more dispersed overall distribution. Applying spatial constraints alone (w/o deep manifold alignment) pulls the residuals closer to the origin; however, the KDE contours reveal an amorphous, diffuse distribution lacking distinct internal organization, mirroring its significant structural divergence (KL 0.1092). In contrast, the variant without RPM transforms the chaotic scatter into a structured bimodal pattern, indicating that deep alignment constraints effectively disentangle latent semantics (lowering KL to 0.0603). Finally, the proposed DSSC-MAE refines this duality. The clusters become more distinct and densely concentrated toward the origin. This confirms that our full strategy not only minimizes the residual magnitude but also preserves a rigorous internal topology, where diverse semantic categories are organized into coherent, separable regimes rather than collapsed into a chaotic mean.

2) Ablation on Mechanisms Mitigating Manifold Misalignment and Trajectory Distortion: As introduced in Section III-A, the inherent domain gap in cross-sensor imagery causes standard tokenizers to produce misaligned latent manifolds. This structural misalignment forces the ODE trajectories to curve, thus hindering consistency training stability and single-step inference performance. To quantitatively evaluate how the proposed framework specifically aligns these manifolds and reduces trajectory curvature, we systematically ablate the manifold alignment components and the generative training configurations. As detailed in Table V, we evaluate this synergy using three metrics. Central moment discrepancy (CMD) [71] measures the macroscopic global domain gap between latent manifolds. Local mutual information (LMI) [72], computed via soft-probabilistic matching, evaluates microscopic feature similarity across manifolds.

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TABLE IV

QUANTITATIVE ABLATION OF THE RPM AND HIERARCHICAL ALIGNMENT STRATEGY. WHILE SHALLOW ALIGNMENT YIELDS HIGH PSNR, THE PROPOSED DEEP MANIFOLD ALIGNMENT IS CRUCIAL FOR MINIMIZING STRUCTURAL DIVERGENCE (KL) AND OPTIMIZING PERCEPTUAL QUALITY (LPIPS/FID)

Method	Reconstruction Quality			Residual Gap Analysis	
	PSNR ↑	LPIPS ↓	FID ↓	KL (Structure) ↓	ℓ_2 (Magnitude) ↓
Stage I MAE (w/o Stage II)	26.86	0.6866	251.92	0.1990	221.62 ± 38.67
DSSC-MAE w/o Sinkhorn OT	30.66	0.4040	111.29	0.1521	193.67 ± 30.67
DSSC-MAE w/o Patch-wise NCE	30.84	0.4178	110.45	0.0650	157.44 ± 27.24
DSSC-MAE w/o Deep Manifold Alignment	**30.85**	0.4332	120.58	0.1092	176.29 ± 32.49
DSSC-MAE w/o RPM (Linear Embedding)	30.65	0.3971	109.68	0.0603	164.49 ± 23.87
Proposed DSSC-MAE	30.81	**0.3585**	**99.24**	**0.0589**	**155.40 ± 26.58**

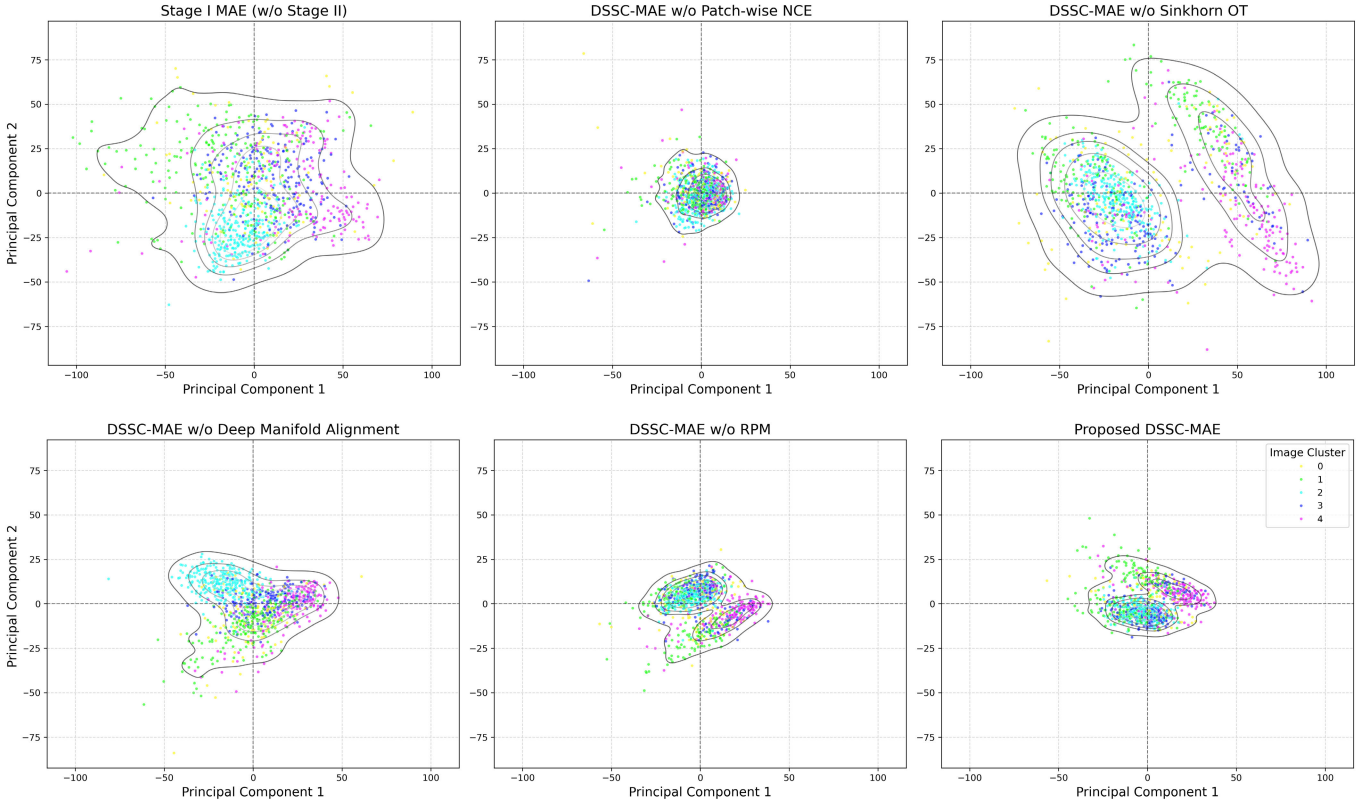


Fig. 10. PCA visualization of latent residual vectors ($\mathbf{z}_{\text{HR}} - \mathbf{z}_{\text{LR}}$) color-coded by CLIP-derived semantic clusters. The Stage I MAE shows high-variance dispersion. The variant w/o patch-wise NCE minimizes the overall magnitude through global OT, while the variant w/o sinkhorn OT yields richer local semantic separations via NCE but remains globally dispersed. The proposed DSSC-MAE optimally harmonizes both constraints, yielding distinct and tightly concentrated clusters near the origin to preserve manifold topology.

Finally, trajectory curvature [26] measures the deviation of the generated ODE trajectory from the ideal linear path.

As shown in Table V, the KL-VAE baseline exhibits inherent limitations in cross-sensor scenarios. Specifically, its latent manifolds demonstrate low microscopic feature similarity, reflected in a low LMI of 0.0813. Coupled with a severe macroscopic global domain gap indicated by a CMD of 32.06, this structural misalignment contributes to an elevated ODE trajectory curvature of 0.4287. Similarly, while the unaligned MAE baseline prioritizes semantic representations to reduce the macroscopic gap to a CMD of 16.48, it struggles to align microscopic features due to the lack of an explicit manifold alignment mechanism, resulting in a comparably low LMI of 0.0851.

To clarify the relationship between manifold geometry (CMD and LMI) and the ODE trajectory curvature, we first isolate the contribution of the generative configuration. On the unaligned KL-VAE manifold with a CMD of 32.06, replacing the standard residual CM (RCM) with the proposed LSIRCM reduces the curvature from 0.4287 to 0.3440. This demonstrates the independent stabilizing effect of the AnchorDiT architecture and the noise-aware hybrid consistency training. However, the trajectory remains curved, indicating that generative improvements alone cannot fully overcome the geometric bottleneck imposed by misaligned manifolds.

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TABLE V

SUBTRACTIVE ABLATION ISOLATING THE MECHANISMS MITIGATING CROSS-SENSOR MANIFOLD MISALIGNMENT AND GENERATIVE TRAJECTORY DISTORTION. EFFECTIVE MANIFOLD ALIGNMENT AIMS TO SIMULTANEOUSLY MINIMIZE THE GLOBAL GAP CMD AND PRESERVE LMI

Tokenizer Type	Generator Type	CMD ↓	LMI ↑	Curvature ↓	PSNR ↑	FID ↓
KL-VAE Baseline	Standard RCM	32.06	0.0813	0.4287	25.40	95.29
KL-VAE Baseline	LSIRCM	32.06	0.0813	0.3440	27.21	73.15
MAE (w/o Alignment)	LSIRCM	16.48	0.0851	0.2997	28.28	51.78
DSSC-MAE w/o Patch-wise NCE	LSIRCM	11.43	0.1377	0.2245	29.59	43.19
DSSC-MAE w/o Sinkhorn OT	LSIRCM	14.65	0.1860	0.1867	29.46	38.55
Proposed DSSC-MAE	LSIRCM	**11.27**	**0.1880**	**0.1544**	**29.72**	**30.98**

TABLE VI

ABLATION OF THE PREDICTION TARGET. THE PROPOSED RESIDUAL FORMULATION ($\Delta\mathbf{z}$) YIELDS SUPERIOR STRUCTURAL AND PERCEPTUAL FIDELITY COMPARED TO DIRECT PREDICTION ($\mathbf{z}_{\text{HR}}$)

Prediction Target	PSNR ↑	LPIPS ↓	FID ↓
$\mathbf{z}_{\text{HR}}$ (Direct Prediction)	29.64	0.2157	37.52
$\Delta\mathbf{z}$ (Residual Formulation)	**29.72**	**0.2140**	**30.98**

By fixing the generator to LSIRCM, we investigate how specific manifold geometry (CMD and LMI) influences the ODE trajectory. Operating on the unaligned MAE reduces the macroscopic gap to a CMD of 16.48, which correspondingly decreases the curvature to 0.2997, suggesting that a closer global distribution shortens the transport trajectory. Furthermore, employing only Sinkhorn OT without patch-wise NCE further narrows the macroscopic gap to a CMD of 11.43, dropping the curvature to 0.2245. Conversely, relying solely on patch-wise NCE without Sinkhorn OT prioritizes microscopic feature similarity, yielding a higher LMI of 0.1860. This local structural preservation mitigates transport trajectory distortion, significantly reducing the curvature to 0.1867 despite a broader global gap with a CMD of 14.65. The proposed DSSC-MAE integrates both mechanisms to establish a refined topological basis that is simultaneously aligned macroscopically with a CMD of 11.27 and microscopically with an LMI of 0.1880. Based on this sensor-invariant manifold, LSIRCM achieves a more linearized trajectory with a curvature of 0.1544. Therefore, this straightened ODE flow mitigates truncation errors during single-step inference, yielding superior structural fidelity with a PSNR of 29.72 dB and improved perceptual quality with an FID of 30.98.

3) Effectiveness of Controllable Residual Generation: To validate our design choice in Phase 2, we assessed the benefit of the controllable residual generation strategy. We compared our approach, which models the residual difference ($\Delta\mathbf{z}$), against a baseline that directly predicts the absolute HR latent target ($\mathbf{z}_{\text{HR}}$).

Table VI presents the comparison. The direct prediction method faces a more difficult optimization challenge, as it must reconstruct the entire signal distribution from scratch. This increased complexity leads to suboptimal performance, yielding a PSNR of 29.64 dB and an FID of 37.52. In contrast, our residual formulation yields superior results, improving PSNR to 29.72 dB and LPIPS to 0.2140 while significantly reducing FID to 30.98.

TABLE VII

SENSITIVITY ANALYSIS OF THE WEIGHTING PARAMETER γ. THE UNWEIGHTED BASELINE ($\gamma = 0$) SUFFERS FROM HIGH PERCEPTUAL ERROR, WHILE OUR STRATEGY ($\gamma = 12$) EFFECTIVELY PREVENTS ARTIFACT GENERATION

Weighting Factor γ	PSNR ↑	LPIPS ↓	FID ↓
0.0 (No Weighting)	**29.77**	0.2188	43.18
6.0	29.73	0.2151	35.74
12.0 (Ours)	29.72	**0.2140**	**30.98**
18.0	29.68	0.2158	32.77

The success of the residual approach stems from its ability to use the aligned latent priors ($\mathbf{z}_{\text{LR}}$) as structural anchors. By simplifying the task to learning a zero-centered, low-variance distribution, the model effectively stabilizes the consistency training, focusing solely on synthesizing missing high-frequency details.

4) Sensitivity to Noise-Aware Weighting: To verify the impact of the noise-aware weighting strategy within our hybrid consistency training, we conducted a sensitivity analysis on the decay hyperparameter γ. This parameter modulates the regression loss magnitude across varying noise levels, explicitly controlling the model's attentional focus during the generation process.

Table VII presents the performance variations under different γ settings. The baseline setting of $\gamma = 0.0$ is equivalent to disabling the strategy, applying uniform weighting regardless of noise intensity. While this configuration yields the highest PSNR (29.77 dB), it results in a significant degradation in perceptual quality, with the FID spiking to 43.18 and LPIPS rising to 0.2188. This discrepancy reveals a critical issue. Without adaptive suppression, the regression objective in high-noise states forces the model to overfit to the intense Gaussian noise rather than the underlying signal, resulting in severe artifact generation.

In contrast, increasing γ activates the adaptive modulation mechanism. Empirically, $\gamma = 12$ achieves the optimal tradeoff, significantly reducing FID to 30.98 and LPIPS to 0.2140 while maintaining a competitive PSNR of 29.72 dB. This demonstrates that by downweighting the regression loss in high-noise states, our strategy effectively filters out disruptive gradients, allowing the model to focus on structural alignment only when the signal-to-noise ratio is favorable. However, excessively increasing γ (e.g., to 18.0) leads to diminishing

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TABLE VIII

ABLATION OF CONDITIONAL INJECTION MECHANISMS. THE BASELINE (W/O CROSS-ATTENTION) RELIES ON INPUT CONCATENATION. THE PROPOSED METHOD (OURS) INTEGRATES CROSS-ATTENTION TO PROVIDE SUPERIOR STRUCTURAL GUIDANCE

Method	PSNR ↑	LPIPS ↓	FID ↓
w/o Cross-Attention	29.69	0.2144	36.97
Ours	**29.72**	**0.2140**	**30.98**

TABLE IX

ABLATION STUDY ON INFERENCE STEPS. LSIRCM ACHIEVES ROBUST CONVERGENCE IN A SINGLE STEP, WITH ADDITIONAL STEPS YIELDING NEGLIGIBLE PERCEPTUAL GAINS

Inference Steps	PSNR ↑	LPIPS ↓	FID ↓
1 (Ours)	**29.72**	0.2140	30.98
2	29.70	0.2217	30.21
4	29.65	**0.2137**	**28.23**

returns, as the regression guidance becomes too weak to constrain the solution space effectively.

5) Evaluation of Conditional Injection Strategy: To validate the architectural advantage of the conditional injection design, we isolated the specific contribution of the cross-attention mechanism. We compared the full model against a baseline variant denoted as "w/o Cross-Attention," which relies exclusively on input concatenation to inject the structural prior $\mathbf{z}_{\mathrm{LR}}$.

As reported in Table VIII, while the concatenation-based baseline achieves a comparable pixel-wise metric (PSNR 29.69 dB), it yields significantly inferior perceptual performance (FID 36.97). This discrepancy suggests that rigid spatial concatenation tends to entangle structural guidance with noisy latent features, limiting the model's ability to generate realistic textures.

Conversely, our proposed configuration integrates cross-attention to achieve notable improvements, particularly in generative realism (lowering FID to 30.98). By treating the sensor-invariant prior as a dynamic reference distinct from the noisy input stream, the model can adaptively query relevant structural cues. This indicates that complementing input concatenation with cross-attention is beneficial for enhancing perceptual fidelity in cross-sensor reconstruction.

6) Impact of Sampling Steps on Restoration Quality: To evaluate the single-step inference protocol, we investigate the tradeoff between sampling steps and restoration fidelity. Because the aligned latent manifold inherently lowers the ODE trajectory's curvature and minimizes truncation errors, LSIRCM leverages hybrid consistency training to condense the sampling trajectory into a direct, deterministic mapping.

Table IX presents the quantitative analysis across varying inference steps. The results empirically demonstrate that LSIRCM achieves optimal structural convergence in a single forward pass. Notably, the single-step inference yields the highest structural fidelity (PSNR: 29.72 dB), surpassing multistep configurations. While extending the inference to two or four steps provides marginal gains in perceptual metrics (lowering FID from 30.98 to 28.23), it incurs a slight degradation in pixel-wise accuracy PSNR and, fundamentally, imposes a linear increase in computational latency.

This stability corroborates that our model effectively establishes a robust mapping from the prior to the target distribution. Since single-step inference already guarantees high fidelity, sacrificing computational speed for negligible perceptual improvements lacks practical justification. Consequently, this configuration achieves an optimal efficiency–quality tradeoff for time-critical remote sensing applications.

TABLE X

BENCHMARK OF INFERENCE EFFICIENCY AND RECONSTRUCTION QUALITY. TIME IS REPORTED AS THE AVERAGE DURATION PER 256 × 256 PATCH ON AN NVIDIA A100 GPU

Method	Steps	Avg. Time (ms/patch) ↓		Quality Metrics	
		BS=1	BS=16 (Amortized)	PSNR ↑	FID ↓
SR3	2000	37923.77	19681.74	18.13	79.27
EDiffSR	100	1995.07	1747.84	27.49	62.86
LDM	50	341.62	256.15	20.57	39.37
FastDiffSR	20	293.99	185.24	24.86	52.57
LSIRCM (Ours)	**1**	**59.85**	**14.50**	**29.72**	**30.98**

7) Inference Efficiency and Model Complexity: For diffusion-based models to be viable in operational Earth observation, they must overcome the bottleneck of inference latency. To validate the practical superiority of LSIRCM, we benchmarked its efficiency against representative baselines on a high-performance computing platform (NVIDIA A100 GPU). We evaluated two scenarios: real-time response (Batch Size = 1) and high-throughput batch processing (Batch Size = 16), focusing on the average inference time per 256 × 256 patch.

Table X details the computational benchmarks. Traditional pixel-space diffusion models incur substantial costs due to iterative sampling. Similarly, standard latent models like LDM still necessitate 50 sampling steps, consuming 256.15 ms per patch. EDiffSR requires over 1700 ms per patch, and even the latent-based FastDiffSR (185.24 ms) remains a limiting factor for high-volume data streams. In contrast, LSIRCM demonstrates a transformative efficiency gain. By leveraging a deterministic single-step mapping, the amortized inference time is reduced to 14.50 ms per patch. This represents a speedup of approximately 13× over FastDiffSR, 120× against EDiffSR, nearly 18× over LDM, and 1357× compared to SR3 all while maintaining superior structural fidelity.

To further demonstrate the value of LSIRCM in operational workflows, we visualized the estimated processing time for a standard remote sensing production task: reconstructing a full-scene image from the Gaofen-6 (GF-6) satellite (67 000 × 54 000 pixels, approx. 55 200 patches). As illustrated in Fig. 11, the logarithmic scale reveals a stark contrast in operational feasibility. While SR3 would require nearly two weeks (approx. 18 402 min, extended red bar) and EDiffSR over a day to process a single scene, and LDM still requires nearly 4 h (235.66 min), LSIRCM completes the entire reconstruction in just 13.34 min. This capability transforms SR from a computationally expensive offline task into a feasible component of daily production pipelines, enabling near real-time processing for time-critical applications such as disaster response and crop monitoring.

The training costs (Table XI) further confirm these performance gains in practical applications. Although lightweight

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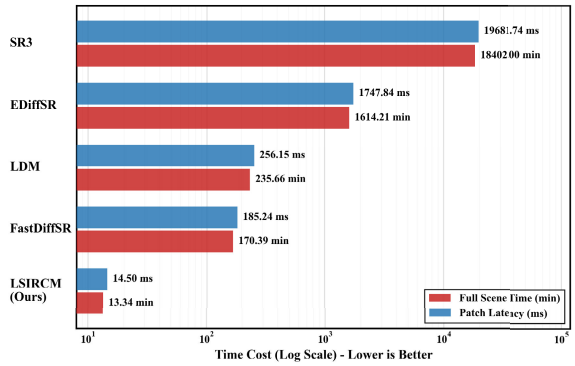


Fig. 11. Logarithmic comparison of inference efficiency and operational feasibility. The shorter bars for LSIRCM (bottom) illustrate an order-of-magnitude acceleration in both patch latency (blue) and full-scene processing time (red), validating its suitability for time-critical remote sensing applications.

TABLE XI

COMPARISON OF MODEL COMPLEXITY AND TRAINING COSTS. TRAINING TIME AND PEAK VRAM WERE MEASURED ON A SINGLE NVIDIA A100 GPU. FOR LATENT MODELS, THE PARAMETER COUNT REPRESENTS THE GENERATIVE BACKBONE

Method	Params (M)	Peak VRAM (GB)	Total Train Time (Days) ↓
SR3	97.81	37.7	5.65
EDiffSR	20.41	32.2	7.40
LDM	275.51	33.1	5.91
FastDiffSR	23.86	35.0	4.71
LSIRCM (Ours)	**158.85**	**26.1**	**3.99**

models like EDiffSR and FastDiffSR deliberately reduce the number of parameters, they still incur high training time costs (up to 7.40 days) due to their operation in the high-dimensional pixel domain. Conversely, while standard latent models like LDM have larger capacities, explaining their long training times of up to 5.91 days, their performance is limited by the bottleneck of the shared tokenizer. Encoding heterogeneous inputs using a single tokenizer leads to latent manifold misalignment, which complicates the ODE trajectory and reduces the quality of fast sampling, even with significant computational investment.

This significant efficiency gain is attributed to the synergy between our decoupled sensor-invariant manifold learning, hybrid consistency training, and the AnchorDiT architecture. By explicitly compressing and aligning heterogeneous data, this preliminary stage straightens the generative ODE trajectory. The consistency training strategy then streamlines the iterative probability flow into a direct mapping for speed, while the AnchorDiT utilizes aligned latent priors to enforce fidelity. This structurally simplified probability flow significantly reduces the overall optimization complexity, achieving a highly efficient total training time of 3.99 days. This dual mechanism ensures robustness in rapid single-step inference, positioning LSIRCM as a highly promising solution for large-scale remote sensing applications.

V. CONCLUSION

In this article, we present a unified framework for cross-sensor remote sensing SR that effectively addresses the tradeoff between inference efficiency and generative realism. By explicitly decoupling sensor-invariant manifold learning from generative reconstruction, we demonstrate that robust cross-sensor manifold alignment is essential for mitigating the curvature of generative ODE trajectories, thereby facilitating efficient high-fidelity reconstruction in latent space. Specifically, the proposed DSSC-MAE enforces alignment via a hierarchical strategy that synergizes global Sinkhorn OT with local patch-wise contrastive constraints to construct a compressed sensor-invariant latent space. Leveraging this aligned prior to mitigate cross-sensor domain shifts, LSIRCM employs a structurally anchored strategy that integrates with Hybrid Consistency Training to transform the iterative sampling trajectory into a deterministic single-step mapping without compromising perceptual fidelity. Extensive quantitative and qualitative evaluations on cross-sensor datasets demonstrate that our framework achieves superior structural fidelity, surpassing the visual realism of state-of-the-art diffusion models while maintaining the inference speed of regression networks.

Beyond its methodological advancements, this work offers a practical pathway for operational Earth observation. By effectively bridging the gap between advanced generative capability and real-time deployment standards, it enables the integration of high-quality SR into time-sensitive and large-scale monitoring workflows. Nevertheless, the current implementation is primarily optimized for fixed sensor pairs, which constrains immediate generalization to arbitrary modalities. Future research will focus on evolving this decoupled alignment strategy into a generalist remote sensing foundation model, leveraging large-scale datasets to enable seamless adaptation across diverse sensor combinations.

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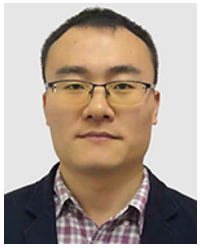
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